Dell PowerFlex 4.5.x and PowerFlex Manager 4.8.x

Software Administration Guide

August 2026
Rev. 1.3

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This document provides procedures for using Dell PowerFlex software through the PowerFlex Manager user interface or PowerFlex SCLI.

The target audience for this document includes system administrators responsible for managing PowerFlex software through the PowerFlex Manager user interface or PowerFlex SCLI.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
management virtual machine	MVM
management data store	MDS
metadata manager	MDM
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data server	SDS
storage virtual machine	SVM
storage data replicator	SDR
storage data target	SDT
system configuration reporter	SCR

The following table summarizes key terminology changes. References in the documentation will be updated over time.

Table 2. Terminology changes

Term	Formerly known as
PowerFlex management cluster (ESXi)	PowerFlex management controller 2.0
PowerFlex management cluster (vSAN)	PowerFlex management controller 1.0
PowerFlex management cluster	PowerFlex management controller
Single ESXi PowerFlex management node	PowerFlex management node
PowerFlex management node	PowerFlex controller node
Dell Technologies connectivity services	SupportAssist

Table 2. Terminology changes (continued)

Term	Formerly known as
Enterprise Project Services (EPS) form	Logical Configuration Survey (LCS)
Dell Code and Compatibility Knowledge Hub	Release Certification Matrix (RCM) Portal

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 3. Revisions

Date	Document revision	Description of changes
August 2026	1.3	Editorial changes
June 2026	1.2	Updated instructions for maintaining nodes in an integrated PowerFlex management (non-KVM) deployment.
April 2026	1.1	Editorial updates
November 2025	1.0	Initial release

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Completing tasks in PowerFlex Manager

This section specifies the location of tasks that you perform through PowerFlex Manager. The topic lists the tasks and corresponding location in the Dell PowerFlex Manager User Guide.

Table 4. Location of task in the Dell PowerFlex Manager User Guide

If you want to....	Go here in the Dell PowerFlex Manager User Guide
Acknowledge an alert	Alerts
Add certificates for security	Security
Back up and restore	Serviceability
Complete the initial configuration and setup of your system	Initial configuration and setup
Configure events and alerts	Events and alerts
Create credentials	Security
Create users, LDAP users, directory services	User management
Define networks	Networking
Deploy and manage system resources	Deploying a resource configuration in a resource group
Deploy PowerFlex using a CSV file	Deploy a PowerFlex cluster using a CSV topology file
Discover resources	Discover a resource
Discovery overview	Resources
Generate a troubleshooting bundle	Serviceability
Get started with PowerFlex Manager	Getting started
Lifecycle management of resource groups	Lifecycle
Manage block storage	Managing block storage after PowerFlex deployment
Manage resource groups	Additional resource group management tasks
Manage resources	Additional resource management tasks
Manage templates	Additional template management tasks
Manage templates	Templates
Managing system data networks	Managing system data networks in PowerFlex Manager
Monitor scheduled jobs	Jobs
Network types	Settings
Protect your block environment	Protecting a block storage environment
Remove a device	Remove devices
Update a PowerFlex or other license	License management
Update passwords	Updating passwords for system components
Upgrade software	Software upgrade
Use templates to deploy your system	Preparing a template for a resource group deployment
View and interpret the graphs on the dashboard	Displaying system information at a glance

Table 4. Location of task in the Dell PowerFlex Manager User Guide (continued)

If you want to....	Go here in the Dell PowerFlex Manager User Guide
View an event	Monitor events
Working with repositories and compliance	Repositories

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Expanding PowerFlex storage

This section includes procedures to add PowerFlex nodes or add features to PowerFlex by editing an existing CSV file. It also contains a procedure on how to deploy or expand a PowerFlex cluster using REST API commands.

Expand PowerFlex using a CSV file

Take the following into consideration:

- While expanding nodes with SDTs, you may see errors listed in the **Events** tab. These errors are for the existing components, such as SDTs in the cluster.
- If SSH keys are used, ensure that you use a single SSH key for all nodes when expanding a PowerFlex cluster using a CSV topology file.
- Download and decompress the `Software_Only_Complete_4.x.x.zip` file from the Dell Technologies support site.

Adding new components or features by editing the existing CSV file

Use PowerFlex systems existing CSV topology file to add components or new features to your system.

The CSV file that is used to deploy a system can be used later, in the following scenarios, for adding features or components to an existing PowerFlex system:

- Add SDSs, with or without new storage and acceleration devices
- Add MDMs or tiebreakers. The PowerFlex installer adds them to the MDM cluster and configures them automatically.
- Add SDRs for the replication feature
- Add SDCs
- Add SDTs
- Configure storage pools
- Configure PowerFlex system features which can be configured using the CSV file, such as performance profiles, acceleration, and so on.

To add and configure components and features, make a copy of your system's latest CSV topology file. Add rows or columns, or edit existing rows and columns. For new features, you can copy columns from the latest complete CSV template, and paste them into the copy of your system's CSV file. Get the latest complete template, see related information.

The following examples show how you can edit the CSV topology file which can be edited for various system modification scenarios.

Add storage and acceleration:

Figure 1. Add storage and acceleration to the CSV topology file

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Username	Passwor	Operat	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Prote	Fault Set	SDS Storage Device L	StoragePool
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
3	root	Scaleio1	linux	Secondar	MDM_115	10.55.137.	10.55.137	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
4	root	Scaleio1	linux	Secondar	MDM_116	10.55.137.	10.55.137	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb	sp2
7														
8														

Edited CSV file—new rows for added SDSs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Username	Password	Is MDM/TI	MDM Mgn	MDM IPs	MDM Nan	perfProfil	Virtual IPs	Virtual IP	Is SDS	SDS Name	SDS All IPs	SDS-SDS C	SDS-SDC C	Protectior	Fault Set	SDS Stora	StoragePc
2	root	Password	Primary	10.76.1.1	10.76.60.1	MDM1	High	10.76.60.2	eth1	Yes		10.76.60.195			domain1	fs1	/dev/sdb,	pool1
3	non_root(	Password	Secondar	10.76.1.2	10.76.60.1	MDM2	High	10.76.60.2	eth1	Yes		10.76.60.17			domain1	fs1	/dev/sdb,	pool2
4	root	Password	Secondar	10.76.1.3	10.76.60.1	MDM3	High	10.76.60.2	eth1	Yes		10.76.60.18			domain1	fs2	/dev/sdb,	pool1,poo
5	root	Password	TB		10.76.60.3	TB1	High			Yes	SDS1	10.76.60.3	10.76.2.2,	10.76.3.3	domain1	fs3	/dev/sdb,	pool1,poo
6	root	Password	TB		10.76.60.1	TB2	High			No								
7	root	Password	Standby-S	10.76.1.4	10.76.60.2	MDM4	High	10.76.60.2	eth1	No								
8	root	Password	Standby-S	10.76.1.5	10.76.60.1	MDMa	High	10.76.60.2	eth1	Yes		10.76.60.175			domain2		/dev/sdb,	pool1,poo
9	root	Password	Standby-S	10.76.1.6	10.76.60.2	MDMb	High	10.76.60.2	eth1	Yes		10.76.60.22			domain2		/dev/sdb,	pool1,poo
10	root	Password	Standby-TB		10.76.60.3	Tba	High			Yes	SDSa	10.76.60.35			domain2		/dev/sdb,	pool1,poo
11																		

Add an MDM and a tiebreaker:

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Username	Password	Operating	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Stora	StoragePc	Is SDC	Is SDR
2	root	Scaleio12	linux	Primary	MDM_118	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
3	root	Scaleio12	linux	Secondary	MDM_120	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
4	root	Scaleio12	linux	TB	MDM_121	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
5															

Edited CSV file—new rows added for an additional MDM and TB

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Username	Password	Is MDM/TI	MDM Mgn	MDM IPs	MDM Nan	perfProfil	Virtual IPs	Virtual IP	Is SDS	SDS Name	SDS All IPs	SDS-SDS C	SDS-SDC C	Protectior	Fault Set
2	root	Password	Primary	10.76.1.1	10.76.60.1	MDM1	High	10.76.60.2	eth1	Yes		10.76.60.195			domain1	fs1
3	non_root(	Password	Secondar	10.76.1.2	10.76.60.1	MDM2	High	10.76.60.2	eth1	Yes		10.76.60.17			domain1	fs1
4	root	Password	Secondar	10.76.1.3	10.76.60.1	MDM3	High	10.76.60.2	eth1	Yes		10.76.60.18			domain1	fs2
5	root	Password	TB		10.76.60.3	TB1	High			Yes	SDS1	10.76.60.3	10.76.2.2,	10.76.3.3	domain1	fs3
6	root	Password	TB		10.76.60.1	TB2	High			No						
7	root	Password	Standby-S	10.76.1.4	10.76.60.2	MDM4	High	10.76.60.2	eth1	No						
8	root	Password	Standby-S	10.76.1.5	10.76.60.1	MDMa	High	10.76.60.2	eth1	Yes		10.76.60.175			domain2	
9	root	Password	Standby-S	10.76.1.6	10.76.60.2	MDMb	High	10.76.60.2	eth1	Yes		10.76.60.22			domain2	
10	root	Password	Standby-TB		10.76.60.3	Tba	High			Yes	SDSa	10.76.60.35			domain2	
11																

Figure 2. Add an MDM and a tiebreaker to the CSV topology file

Add storage data replicators (SDRs) for replication:

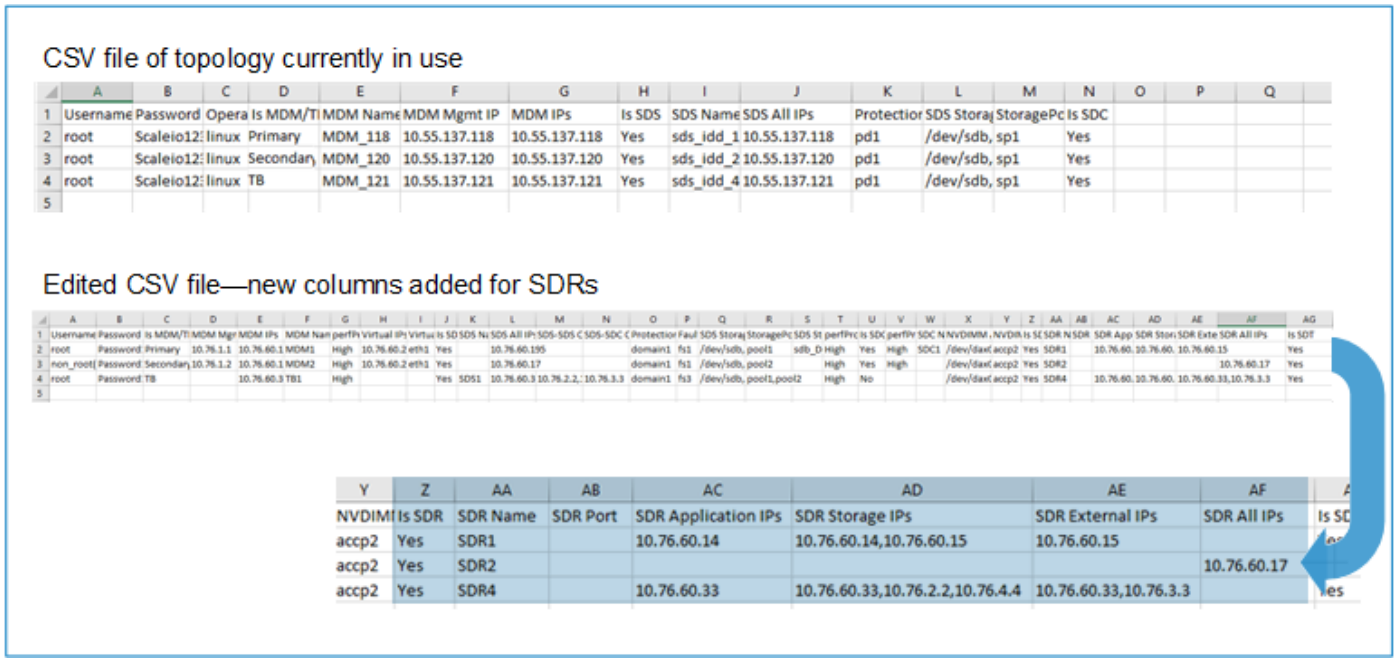


Figure 3. Add storage data replicators to the CSV topology file

Add storage data clients (SDCs):

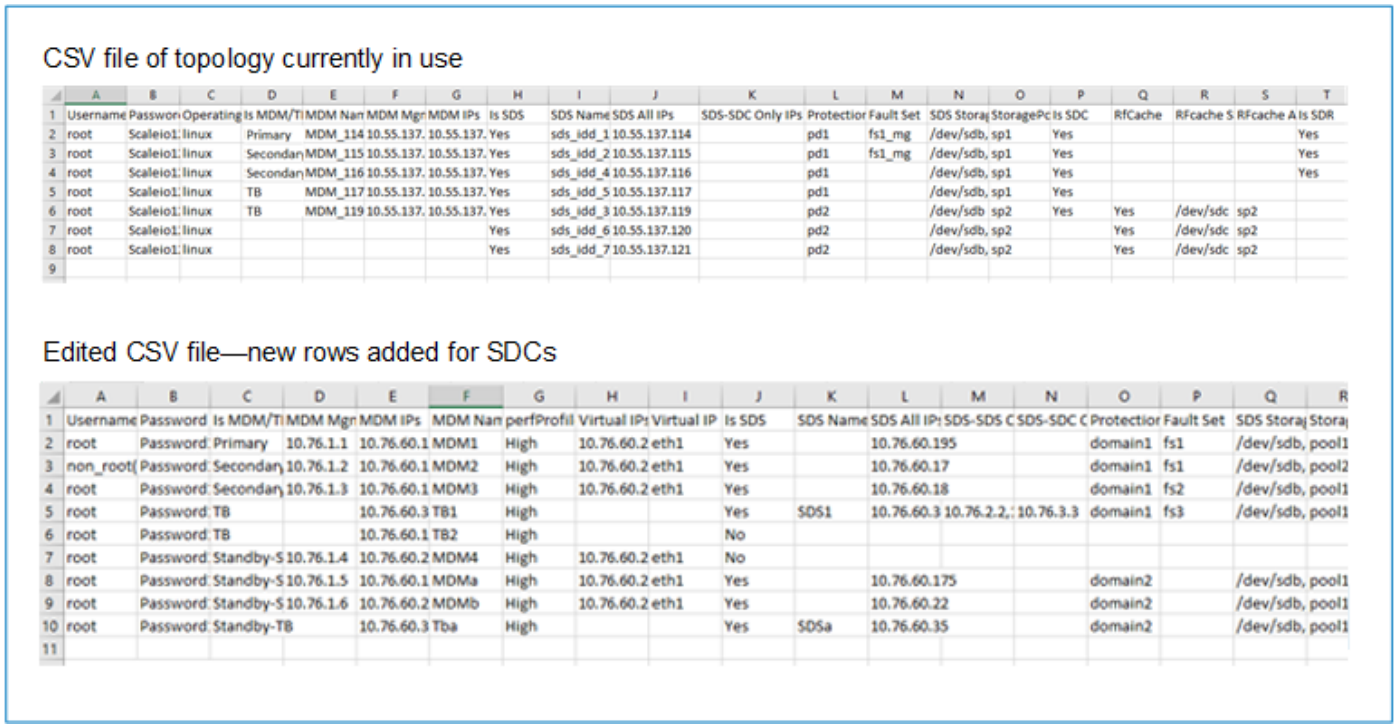


Figure 4. Add SDCs to the CSV topology file

Configure storage pools:

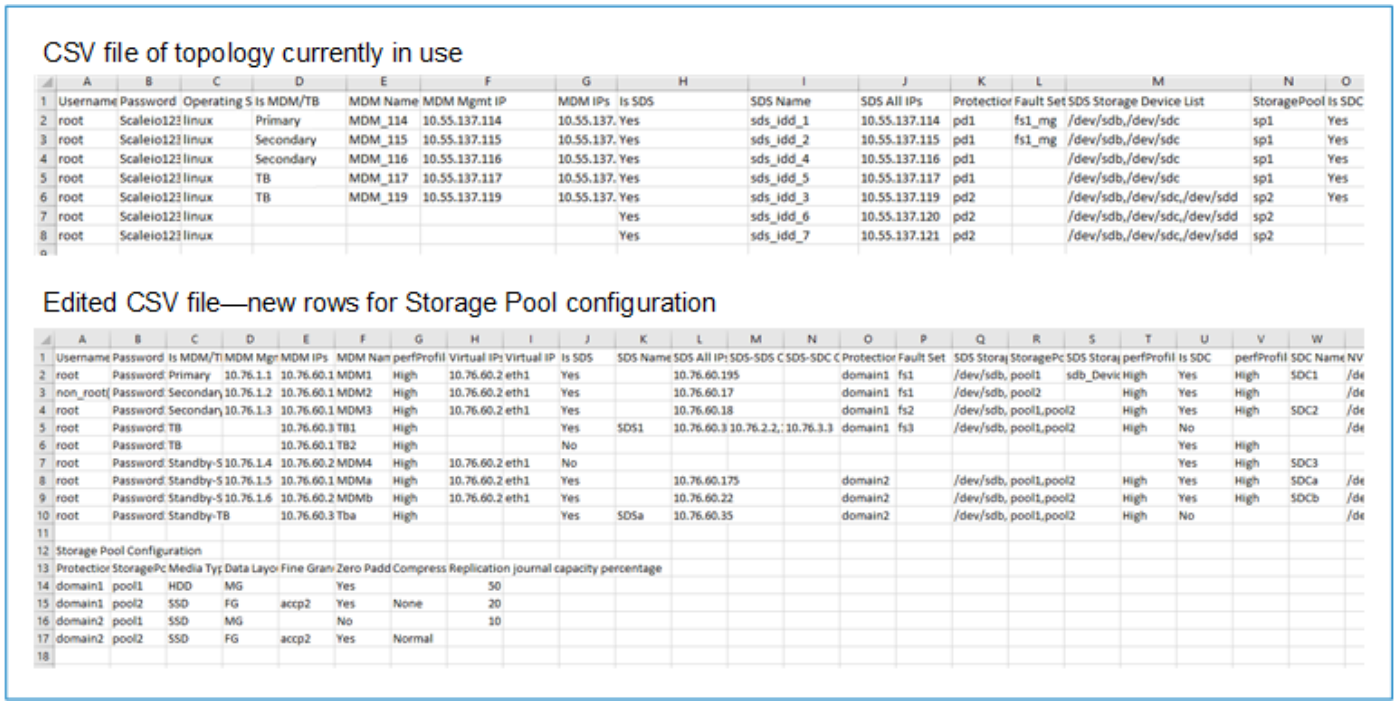


Figure 5. Configure storage pools in the CSV topology file

Deploy or expand a cluster using REST API commands

After you have prepared the CSV topology file, use the REST API to parse it and upload the configuration to PowerFlex Manager, and then deploy block storage resources using REST API commands. This procedure is only recommended for advanced users familiar with working with the REST API.

- Ensure that you have prepared the CSV topology file.
- Ensure that you know the load balancer IP address and user credentials of PowerFlex Manager. All the steps performed in this procedure must be performed using the load balancer IP address. The load balancer IP address is the same IP address you are using to access PowerFlex Manager.
- Ensure that you have the software packages of the PowerFlex core components that you want to deploy, and the ActiveMQ package.
- See the Dell Technologies Developer portal for information about working with the REST API.
- Log in using the REST API, and make a note of the Keycloak token that was returned from `/auth/login` or from `/api/gatewayLogin`.

The following procedure provides examples using cURL.

1. Get the configuration from the CSV file, by parsing it using the PowerFlex REST API. Use the command `POST /im/types/Configuration/instances/actions/parseFromCSV`, where the body is the CSV file reference. For example:

```
curl -v -k -X POST -c /tmp/gw-cookie.txt -H "ContentType:multipart/form-
data" -H "Authorization: Bearer ${token}" -F "file=@<PATH_TO_CSV_FILE>" https://
<IP_ADDRESS>/im/types/Configuration/instances/actions/parseFromCSV
```

   Where: *{token}* is the Keycloak token, *<PATH_TO_CSV_FILE>* is the path to the CSV topology file, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.

   NOTE: The `-c /tmp/gw-cookie.txt` generates a file to ensure that the same gateway pod is used because here are multiple pods available.

   The command returns a JSON file containing the required cluster configuration. The installation command requires this file in order to install resources. Save this file.

2. Upload the RPM packages to the gateway:

```
curl -k -X POST -b /tmp/gw-cookie.txt -H "Content-Type:multipart/form-data"
-H "Authorization: Bearer ${token}" --form "files=@<path to file>/<lia_rpm>" --
form "files=@<path to file>/<sds_rpm>" --form "files=@<path to file>/<mdm_rpm>"
--form "files=@<path to file>/<activemq_rpm>" https://<ingress_hostname >/im/types/
installationPackages/instances/actions/uploadPackages
```

NOTE: The `-b /tmp/gw-cookie.txt` option ensures that this API call is routed to the same gateway pod as the previous calls.

3. Install block storage resources using the PowerFlex REST API. Use the command `POST /im/types/Configuration/actions/install`, where the body is the configuration that was returned from parsing the CSV file.

   If the parameter *systemId* (type string) is added to the configuration, the cluster is created with the given ID. Otherwise, the gateway generates an ID.

   Optional parameters:

   - noUpload—Skip upload package phase (true or false, default if false)
   - noInstall—Skip install package phase (true or false, default if false)
   - noConfigure—Skip configure phase and repository files (true or false, default if false)
   - noSecurityBootstrap—Skip security bootstrap (part of install phase) (true or false, default if true)
   - noLinuxDevValidation (true or false, default if true)
   - sdsInstances (number, optional)
   - extend—extend an existing system (true or false, default if false)
   - globalZeroPadPolicy (true or false, default if true)
   - forceDeviceTakeover—Clean devices before installing SDS (without it, installation will fail when it runs after uninstall; must not be used in productioni) (true or false, optional)

   For example:

```
curl -v -s -k -i -X POST -b /tmp/gw-cookie.txt -H "Content-Type:application/json" -H
"Authorization: Bearer ${token}" "https://<IP_ADDRESS>/im/types/Configuration/actions/
install?noSecurityBootstrap=false&forceDeviceTakeover=true" -d @<PATH>/config.json
```

   Where: *{token}* is the Keycloak token, *<IP_ADDRESS>* is the PowerFlex Manager IP address, and *<PATH>* is the path to the JSON configuration file that was output in step 1.

   NOTE: The `-b /tmp/gw-cookie.txt` option ensures that this API call is routed to the same gateway pod as the previous calls.

4. Installation is asynchronous and consists of four stages: query, upload, install, and configure. Manually monitor and move from one phase to the next one, using the commands `queryPhaseState` and `moveToNextPhase`. You can also monitor the progress of each stage by viewing events in PowerFlex Manager.
   The following is an example of an installation event:

```
COMMAND_STARTED_ON_OPERATION_INSTALL_AND_PHASE_QUERY
```

5. After installation is complete, clear the gateway state. Use the command `POST im/types/Command/instances/actions/clear`.

   For example:

```
curl -v -s -k -i -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
${token}" "https://<IP_ADDRESS>/im/types/Command/instances/actions/clear"
```

   Where: *{token}* is the Keycloak token, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.

5

Maintaining nodes in an integrated PowerFlex management (non-KVM) deployment

In an integrated PowerFlex management (non-KVM) deployment, you can move a node in and out of maintenance mode so you can perform administrative maintenance tasks.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

Determine the primary MDM

Use the following procedure to determine if you are working with a primary metadata manager (MDM).

Establish an SSH connection to the node.

1. To determine if the node is a primary MDM, run:

```
scli --query_cluster
```

2. Check the output of the previous command for cluster information.
   - If the command does not return cluster information, you can perform maintenance on the node.
   - If the command returns cluster information, you must switch ownership to the secondary MDM.

   To log in to the PowerFlex cluster, run:

```
scli --login --username admin --management_system_ip <powerflex_manager_IP>
```

   To switch cluster ownership, run:

```
scli --switch_mdm_ownership --new_primary_mdm_name <mdm_name>
```

Related information

Prepare a PowerFlex management node for maintenance

Prepare a PowerFlex management node for maintenance

You must terminate or move the pods running on the PowerFlex node before updating, upgrading, or rebooting a PowerFlex management node.

Ensure that ownership is switched from the primary MDM to the secondary MDM before any maintenance activities are performed on the node.

Before updating, upgrading, or rebooting a PowerFlex management node, the pods running on the PowerFlex node must be properly terminated or moved.

1. Establish an SSH connection to the PowerFlex controller node.
2. Label the node for maintenance mode.

a. To display existing node status and role type, run the following command:

```
kubectl get nodes
```

```
Example output:
NAME            STATUS   ROLES                         AGE   VERSION
node1           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
node2           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
node3           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
```

b. Label the node for maintenance by running the following command:

```
kubectl label node <NAME> cmo.maintenance.mode=true
```

```
kubectl label node node1 cmo.maintenance.mode=true
```

3. Verify if the node is a primary MDM by running the following command:

```
scli --query_cluster
```

Check the command output for cluster information. If the command returns cluster information, run the following commands to switch ownership to the secondary MDM. Otherwise, continue to step 4.

a. Log in to the PowerFlex cluster:

```
scli --login --username admin --management_system_ip <powerflex_manager_IP>
```

b. Run this command to switch cluster ownership:

```
scli --switch_mdm_ownership --new_primary_mdm_name <mdm_name>
```

4. Drain the pods that are running on the node with the following command:

```
kubectl drain <NAME> --ignore-daemonsets --delete-emptydir-data
```

NOTE: The command may take more than a minute to complete, as it is not finished when the prompt returns. The process continues in the background, and you must wait four to five minutes before proceeding.

Enter an SDS node into maintenance mode. See **Related information**.

Related information

Determine the primary MDM
Enter an SDS node into maintenance mode

Enter an SDS node into maintenance mode

You must use PowerFlex Manager to enter the SDS node into maintenance mode.

Use the following procedure for a node in an integrated PowerFlex management (non-KVM) deployment.

1. Log in to PowerFlex Manager.
2. Go to **Block** > **SDSs** and select the checkbox of the SDS you are working on.
3. Click **More Actions** and select **Enter Maintenance Mode**.
4. Select **Instant** or **Protected** maintenance mode.

The node is now ready for maintenance. You can update the node and reboot it. When maintenance is complete, return the node to operation.

Related information

Prepare a PowerFlex management node for maintenance
Exit a management node from maintenance mode
Exit an SDS node from maintenance mode

Exit a management node from maintenance mode

To return the node to operation after maintenance, you must wait for the node to rejoin the cluster before exiting the node from maintenance mode.

1. Establish an SSH connection to the node.
2. Wait for the node to rejoin the cluster. From a different node that is part of the integrated PowerFlex management (non-KVM) deployment, run the following command to return the node to service:

   ```
   kubectl get nodes
   ```

   Continue to wait if the status is anything other than `Ready, SchedulingDisabled`.
3. To bring the node back into service, run the following command:

   ```
   kubectl uncordon <NAME>
   ```

   <NAME> is the name of the node.
4. Check the status by running the following command:

   ```
   kubectl get nodes
   ```

   Wait for the status to report `Ready` before continuing.
5. Optional: If you were also performing maintenance on an SDS node, do the following:
   a. Log in to PowerFlex Manager.
   b. Go to **Block** > **SDSs** and select the checkbox of the SDS you are working on.
   c. Click **More Actions** and select **Exit Maintenance Mode**.
   d. Wait until the PowerFlex management platform status indicator light goes green before continuing.
6. To remove the maintenance mode label from the management node, run the following command:

   ```
   kubectl label node <NAME> cmo.maintenance.mode-
   ```

   For example:

   ```
   kubectl label node node1 cmo.maintenance.mode-
   ```
7. In PowerFlex Manager, go to **Monitoring** > **Management Diagnostics** > **Nodes**.

   The **Nodes** page opens which displays a list of PowerFlex management platform nodes.

   - You can click **Add filter** and enter case-sensitive free text to control the list of nodes displayed.
8. Select the node that you want to view.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.
9. Wait until the PowerFlex management platform status indicators change to green before moving on to the next node.

Related information

Enter an SDS node into maintenance mode
Post-maintenance status check for PowerFlex Manager version 4.8 and above

Exit an SDS node from maintenance mode

To return the node to operation after maintenance, you must wait for the node to rejoin the cluster before exiting the node from maintenance mode.

1. Log in to PowerFlex Manager.
2. Go to **Block** > **SDSs** and select the checkbox of the SDS you are working on.
3. Click **More Actions** and select **Exit Maintenance Mode**.
4. Wait until the PowerFlex management platform status indicator light goes green before moving on to the next node.

Related information

Enter an SDS node into maintenance mode
Post-maintenance status check for PowerFlex Manager version 4.8 and above

Post-maintenance status check for PowerFlex Manager version 4.8 and above

You must verify the readiness of the nodes, pods, and PowerFlex Manager database after maintenance activities.

1. Log in to PowerFlex Manager.
2. Click on **Monitoring** > **Management Diagnostics**.
3. Verify green status on all panels.

Related information

Exit a management node from maintenance mode
Exit an SDS node from maintenance mode

Shutting down or rebooting non-PowerFlex Manager nodes

You can shut down or reboot non-PowerFlex Manager nodes, and then return the nodes to an operational state.

Shut down or reboot a non-PowerFlex Manager node

Prepare the node for a patching or maintenance operation, such as a part replacement, by entering the node into maintenance mode and shutting down/rebooting the node in a graceful fashion.

CAUTION:

- **When performing tasks on a node that require it to be shut down or rebooted, you must gracefully shut down or reboot the node. Do not force the shut down or reboot operations.**
- **Operating system updates and other maintenance activities, like part replacement, require that you gracefully shut down or reboot a node.**
- **Do not use `-f` in the shutdown or reboot command.**

1. When shutting down/rebooting a node that is a primary MDM (manager), it is recommended that you manually switch MDM ownership to a different node:
   a. From the PowerFlex CLI (SCLI), run:

      ```
      scli --query_cluster
      ```

   b. If the IP addresses of the node are included in the `--query_cluster` output, the faulty node has a role of either MDM or tiebreaker, in addition to its SDS role.

If the IP address of the node is located in the primary MDM role, a switch-over action is required.

c. Switch MDM ownership to a different node:

```
scli --switch_mdm_ownership (-new_primary_mdm_id-- <ID> | --new_primary_mdm_ip <IP>
| --new_primary_mdm_name <NAME>)
```

The node remains in the cluster. The cluster will be in degraded mode after it is powered off, until the faulty component or patch operation in the node is fixed and the node is powered back on.

d. Verify that the cluster status shows that the node is not the primary MDM anymore:

```
scli --query_cluster
```

Output similar to the following should appear, with the relevant node configuration and IP addresses for your deployment:

```
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: 9.20.10.100, 9.20.110.100
Primary MDM:
    ID: 0x775afb2a65ef1f02
        IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses:
10.136.215.239, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Version: 2.0.13000
Secondary MDMs:
    ID: 0x5b2e9f273b7af9b0
        IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses:
10.136.215.223, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
    ID: 0x5828f65b15e778f1
        IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses:
10.136.215.232, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
Tiebreakers:
    ID: 0x6618e0b804644ca4
        IP Addresses: 9.20.10.101, 9.20.110.101, Port: 9011
        Status: Normal, Version: 2.0.13000
    ID: 0x12534ccb3d28fee3
        IP Addresses: 9.20.10.103, 9.20.110.103, Port: 9011
        Status: Normal, Version: 2.0.13000
```

In the example output, the primary MDM IP addresses are:

```
IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses: 10.136.215.239
```

The secondary IP addresses are:

```
IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses: 10.136.215.223
IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses: 10.136.215.232
```

2. Move all applications to a different node:
   - On a VMware ESXi node that is not a cluster member, and that is not configured for HA and DRS, migrate the VMs to another VMware ESXi.
   - On a Linux node, migrate the applications (or the VMs, if the node is running a hypervisor).
3. Log in to PowerFlex Manager as an admin user.
4. On the menu bar, click **Block** > **SDSs**.
5. Select the relevant SDS and click **More Actions** > **Enter Maintenance Mode**.
6. In the **Enter SDS <name> into Maintenance Mode** dialog box, select one of the following options:
   - **Instant**
   - **Protected**
7. Click **Enter Maintenance Mode**.
8. In the confirmation message dialog box, click **Dismiss**.
9. If you are applying a patch:
   a. Run the patch.
   b. Reboot the node, if necessary.

Return the node to an operational state

After a maintenance activity, you must return the node to an operational state.

1. Click **Block** > **SDSs**.
2. Select the checkbox for the node in maintenance mode, go to **More Actions**, and select **Exit Maintenance Mode**.
3. Wait until the state of the SDS displays as **Healthy**.
4. Wait for the rebuild and/ or rebalance operations to complete.
   The node is now operational.

Running scripts on hosts

You can run scripts on servers hosting PowerFlex MDM or SDS components. You can use the scripts to update operating systems or firmware. This feature is supported only on Linux-based bare-metal nodes or virtual nodes.

The PowerFlex Installer can run a user-provided script on a node where PowerFlex is deployed. You can use this feature for any purpose that is external to the PowerFlex system, such as running CLI commands or updating an operating system. This feature enables you to run scripts in a secure fashion while maintaining data integrity.

The PowerFlex Installer orchestrates running the script, and ensures that SDSs are placed in maintenance mode to protect data during the process. Parallel script execution is only permitted on SDSs that are in different protection domains. After a script is run on an SDS, the scripts exits the SDS from maintenance mode.

Optionally, you can set the servers to reboot after running the script. You can also run a verification script either after the initial script is run or after the server reboots.

This feature includes the following system validation before running a script:

- The script was uploaded to the node.
- There is no failed storage capacity.
- There is sufficient spare storage capacity.
- The MDM cluster is in a valid state.

Run the script on one host, and prioritize the hosts in the following manner:

1. SDS-only hosts, each time on a single SDS, unless the option `In parallel on different Protection Domains` is enabled
2. Tiebreaker
3. MDMs

> **NOTE:** The script cannot run on a primary MDM. A switch-over MDM command is used before the script runs on a primary MDM. The script cannot be run in parallel on multiple MDMs, or tiebreakers.

Running the script proceeds as follows:

1. The node enters maintenance mode.
2. The script runs on the host.
3. The host reboots, if required.
4. The validation script runs on the node, if required.
5. The node exits maintenance mode.

Run a script on a host

Use this method to run a script on all or some of the Linux-based nodes with an optional reboot and an optional verification script.

> **CAUTION: PowerFlex Manager is required to run a script on a host. A target script is run without validation or verification.**

The script files must be stored in the folder node folder: `/opt/emc/scaleio/lia/bin`. The script file used to run script on the host operation must be named as `patch_script`.

If verification is required, a script to verify the work can be written and named `verification_script`.

The filenames are hard coded and cannot be changed: `patch_script` and `verification_script`. The scripts are required to have these names. Alternatively, they can be uploaded through PowerFlex Manager to the PowerFlex node.

Use the `kubectl cp` command to copy the script file from any of the PowerFlex management platform hosts or VMs to the block-legacy-gateway pods.

For example:

```
kubectl cp -n powerflex patch_script block-legacy-gateway-0:/usr/local/tomcat/temp/kubectl cp -n powerflex patch_script block-legacy-gateway-1:/usr/local/tomcat/temp/
```

The scripts can be either taken from a gateway local folder or downloaded from an HTTP or HTTPS share.

A list of `SdsIds` or `mdmIds` can be provided to explicitly choose the PowerFlex nodes to run on.

API call

```
/im/types/Configuration/actions/liaRunOsPatching
```

NOTE: Before sending the `liaRunOsPatching` call, log in to PowerFlex and get the system configuration. For more information, see the example commands and workflow below.

Required parameters

Either one of the following parameters is mandatory:

- `pdIds`: Run on all PowerFlex nodes that are part of the following protection domains (PD Ids), in decimal format.
- `fsIds`: Run on all PowerFlex nodes that are part of the following fault sets (FS Ids), in decimal format.
- `sdsIds`: Run on all SDSs listed by Ids, in decimal format
- `mdmIds`: Run on all MDMs listed by Ids, in decimal format
- `executeOnAllSdss`: Run on all SDSs (true/false). All nodes running PowerFlex management platform processes are skipped for reboots. The action must be performed manually.
- `executeOnAllMdms`: Run on all MDMs (true/false)

Optional parameters

The following parameters are optional:

- `isRebootRequired`: Indicates if each node reboot is required after running the patch script (values: true/false)

  NOTE: All nodes running PowerFlex management platform processes are skipped for reboots. The action must be performed manually.

- `isVerificationScriptRequired`: Indicates if the verification script is run on each node (values: true/false)
- `isRunningInParallelOnPds`: Indicates if the operation is run in a parallel way on nodes that belong to different PDs (values: true/false)
- `isStopProcessingOnScriptFailure`: Indicates if the entire operation must be stopped if there is a script failure (values: true/false
- `TimeoutMs`: Indicates timeout value for running the patch script in milliseconds
- `isUploadFileNeeded`: Indicates if the gateway upload scripts to the PowerFlex nodes (values: true/false)

  The following fields are relevant when `isUploadFileNeeded` is true:

  - `patchScriptFilePath`: Either the local folder name on the block-legacy-gateway pods or an http/https URL of the patch script
  - `verificationScriptFilePath`: Either the local folder name on the block-legacy-gateway pods or an http/https URL of the verification script
  - `maintenanceModeType`: Maintenance Mode type (values: IMM/PMM)
  - `verificationScriptTimeoutSec`: Verification script timeout in seconds
  - `rebootTimeoutSec`: Node reboot timeout in seconds

Example commands and workflow

The following are the example commands that are used when running a script on a host:

1. Obtain an access token from the PowerFlex Manager instance. The easiest method is to create a shell script that can be sourced to add the proper variables to the user environment

```
INGRESS_IP=<powerflex manager IP>
INGRESS_USER=<powerflex manager user>
INGRESS_PASSWORD=<powerflex manager password>
```

   a. Get a JWT token with `POST /rest/auth/login`.

```
TOKEN=$(curl -s -k --location --request POST "https://${INGRESS_IP}/rest/auth/
login" --header "Accept: application/json" --header "Content-Type: application/
json" --data "{\"username\": \"${INGRESS_USER}\",\"password\": \"${INGRESS_PASSWORD}
\"}")
```

   b. Parse out the access token which is used to call the API and is valid for 5 minutes by default.

```
ACCESS_TOKEN=$(echo "${TOKEN}" | jq -r .access_token)
```

   c. Parse out the refresh token which can be used to get a new JWT token if the access token has expired. It is valid for 30 minutes by default.

```
REFRESH_TOKEN=$(echo "${TOKEN}" | jq -r .refresh_token)
```

   NOTE: The expiration time for access token is five minutes. If required, the above file can be sourced to refresh all variables.

2. Get the JSON of a system configuration, which will be the payload of the patch command. You must replace `liaPassword` and `mdmPassword` manually from null to some string.

   a. Create and save a JSON file like the following, replacing MDM addresses, MDM user, and MDM password with the appropriate values.

```
{
   "mdmIps":["<MDM IP-1>","<MDM-IP2>"],
   "mdmUser":"<mdm user>",
   "mdmPassword":"<mdm password>",
   "securityConfiguration":
   {
      "allowNonSecureCommunicationWithMdm":"true",
      "allowNonSecureCommunicationWithLia":"true",
      "disableNonMgmtComponentsAuth":"false"
   }
}
```

   b. Insert the output of this command (with fixed passwords) into the `config.json` file.

```
curl -s -X POST -k -H "Content-Type: application/json" --data <json file>
-H "Authorization: Bearer ${ACCESS_TOKEN}" https://<powerflex manager IP>/im/types/
Configuration/instances > config.json
```

3. Run the patch command by loading `config.json`:

   NOTE: The `cookiefile` stores session cookies which keep the command flow directed to the same gateway pod.

```
curl -v -k -c cookiefile -X -i POST -H "Content-Type:application/json" -H
"Authorization:
Bearer ${ACCESS_TOKEN}" "https://<powerflex manager IP>/im/types/Configuration/
actions/liaRunOsPatching?executeOnAllSdss=true
&isRebootRequired=true&isPatchScriptRequired=true&isVerificationScriptRequired=true&pa
tchScriptFilePath=”https://
<ipaddress>/patch_script”&verificationScriptFilePath=”https://<ip-address>/
verification_script”&maintenanceModeType=IMM&rebootTimeoutSec=600" -d @config.json
```

4. Query command state by running the following command:

```
curl -k -b cookiefile -X GET -H "Content-Type: application/json" -d @config.json
-H "Authorization: Bearer $ACCESS_TOKEN" "https://10.234.89.250/im/types/ProcessPhase/
actions/queryPhaseState" | jq
{
  "phaseStatus": "idle",
  "phase": "idle",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 0,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}
```

Or:

```
{
  "phaseStatus": "running",
  "phase": "execute",
  "numberOfRunningCommands": 1,
  "numberOfPendingCommands": 1,
  "numberOfCompletedCommands": 35,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}
```

Or:

```
{
  "phaseStatus": "completed",
  "phase": "validate",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 2,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}
```

Look for:

```
{
  "phaseStatus": "completed",
  "phase": "execute",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 37,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}
```

5. Move to the next phase by running the following command:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d ''
-H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250//im/types/ProcessPhase/
actions/moveToNextPhase
```

6. Cancel the phase by running the following command:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d
'' -H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/Command/
instances/actions/abort
```

7. Clear the phase by running the following command:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d
'' -H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/Command/
instances/actions/clear
```

8. Move to idle phase by running the following command:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d ''
-H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/ProcessPhase/
actions/moveToIdlePhase
```

Logs

Gateway pod log locations:

- `/usr/local/tomcat/logs/scaleio.log`
- `/usr/local/tomcat/logs/scaleio-trace.log`

LIA log location: `/opt/emc/scaleio/lia/logs/trc.x`

NOTE: Special switch to keep the script in the node when troubleshooting or testing:

1. Edit the file `/usr/local/tomcat/webapps/ROOT/WEB-INF/classes/gatewayInternal.properties`
2. Find the field `"ospatching.delete.scripts=false"`
3. Change to true for troubleshooting. The default is false.

Fully powering off or fully powering on nodes

You can fully power off or fully power on nodes in an integrated PowerFlex management (non-KVM) deployment. These actions are not part of routine maintenance.

The power off procedure FULLY powers off a PowerFlex system. When you complete the steps in *Power off nodes in an integrated PowerFlex management (non-KVM) deployment*, you lose all access to your PowerFlex cluster.

The power on procedure FULLY powers on a PowerFlex system. Complete the steps in *Power on nodes in an integrated PowerFlex management (non-KVM) deployment* to power on a PowerFlex cluster that is fully powered off.

Power off nodes in an integrated PowerFlex management (non-KVM) deployment

Use the following procedure to fully power off the nodes in an integrated PowerFlex management (non-KVM) deployment.

CAUTION: When you fully power off the PowerFlex system, you lose all access to your PowerFlex cluster.

1. Stop all external workloads. Dell Technologies recommends that you unmount the volumes for the SDCs.
2. Log in to any PowerFlex Manager node using SSH.
3. Stop the PowerFlex Manager database.

   NOTE: Once the PowerFlex Manager database is stopped, the PowerFlex Manager user interface is unavailable, and you cannot authenticate to the MDM cluster.

   a. To gather the IP address of the database service, run these commands:

```
export NAMESPACE=powerflex
```

```
export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o
jsonpath='{.spec.clusterIP}')
```

b. To shut down the database pods and service, run this command:

```
curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
```

c. To check the shutdown status, run this command:

```
curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
```

d. Continue to run the shutdown status command until the command returns a verified status.

The following example output shows a verified status:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

4. Power off all nodes.

NOTE: Nodes running PowerFlex management platform processes may take longer to shut down than simple SDS nodes.

Power on nodes in an integrated PowerFlex management (non-KVM) deployment

Use the following procedure to fully power on the nodes in an integrated PowerFlex management (non-KVM) deployment.

Fully power on the PowerFlex system.

1. Power on all SDS nodes.
2. Log in to the primary MDM using SSH.
3. Validate if the cluster status is normal by running the following command:

```
scli --query_cluster
```

4. Log in to all nodes running PowerFlex management platform processes.
5. From each PowerFlex management platform node, use the `lsblk` command to check the status of the exported volumes, and stop when the partitions are displayed under the `scinia` device.
   If partitions are not visible, but the cluster reports as healthy, use the `partprobe` command to rescan the attached block devices.
6. Run the following command to check the status of the rke2-server process on all the PowerFlex management platform nodes:

```
systemctl status rke2-server
```

   If the status is active, go to the next step. If the status is shown as activating, repeat the status check until active. If status is failed, attempt to start the service by running:

```
systemctl start rke2-server
```

7. Once rke2-server is running, ensure the PowerFlex management platform nodes are in a ready state by running:

```
kubectl get nodes
```

   If a message is displayed, wait for a few minutes and verify again. After nodes are indicated as ready, continue to the next step.
8. Restore the PowerFlex Manager database.

a. To gather the IP address of the database service, run:

```
export NAMESPACE=powerflex
```

```
export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o
jsonpath='{.spec.clusterIP}')
```

b. To start the database pods and service, run:

```
curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"
```

c. To check the start up status, run:

```
curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
```

d. Continue to run the startup status command until the status returns a verified status.

Here is an example output of a verified status:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

9. Log in to PowerFlex Manager:

a. Go to **Monitoring** > **Management Diagnostics** > **Diagnostics**.

The diagnostic page opens which displays PowerFlex management platform diagnostic cards. In the result icon, last run date, run time, and run state are listed for each diagnostic card.

b. To manually start a diagnostic, click **Run Now**.

A notification message confirms that the specified diagnostic has started. After a diagnostic run begins, you can cancel it. If there are errors, a failure message displays. The overview displays the run state, start time, and end time. If errors occur, the diagnostic result displays failure reasons, recommendations to remediate failures, and raw output.

c. Click **View Raw Output** for more information.

6

Administering the CloudLink Center

This section contains procedures to license CloudLink, configure syslog and audit logs in CloudLink, and manage self-encrypting drives in CloudLink.

For more information, see the CloudLink documentation.

Add a license in CloudLink Center

Use this procedure to add licenses to CloudLink Center.

CloudLink license files determine the number of machine instances, CPU sockets, encrypted storage capacity, or physical machines with self-encrypting drives (SEDs) that your organization can manage using CloudLink Center. License files also define the CloudLink Center usage duration.

NOTE: CloudLink center can act as a key management interoperability protocol (KMIP) server if you upload a KMIP license to it.

1. Log in to CloudLink Center.
2. Select **System** > **License**.
3. Click **Upload License**.
4. Browse to select the software or hardware encryption license file and click **Upload**.

   NOTE: If the CloudLink environment is managed by PowerFlex Manager, after you update the license, go to the **Resources** page, select the CloudLink Center VMs, and click **Run Inventory**.

Configure syslog and audit logs in CloudLink

CloudLink supports four multiple syslog configurations from CloudLink 8.2. The audit log and system log are retrieved locally and available for 15 days.

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Add Syslog Server**. The **Add** window appears.
   a. Provide the syslog IP address, port number, and select the facility and protocol. Click **Add**.

      NOTE: For audit logs, choose the facility as audit if the logs are transmitted using PowerFlex Manager.

   b. If the service status of the syslog server is not active, click **Resume**. In the **Confirm Syslog Mode Change** window, click **Resume**.

      The service status is Active.

      NOTE: The **Change Syslog Format** and **Postpone** options can only be done globally.

3. To retrieve the logs locally:
   a. To enable SSH from CloudLink, click **Server** > **Network** > **Change SSH**.
   b. Select the **Enabled** option and click **Change**.
   c. SSH to the CloudLink IP address. The logs are in the `/var/log/CLC_audit.log` file.

7

Managing storage devices using CloudLink

This section describes how to manage the storage devices using CloudLink.

Encrypt the self-encrypting drive or NVMe storage device

CAUTION: Never attempt to encrypt or erase the devices that are currently attached to the SDS. Devices should only be encrypted before they are added to the SDS, or after removing them from the SDS. Any existing data on the device will be destroyed as a result of this procedure.

- You can encrypt SSDs or NVMes using one of the following methods:

Encrypt devices in CloudLink Center

Encrypt the device in CloudLink Center.

1. Click **Agent** > **Machines**.
2. Select the checkbox for the required machine.
3. Click **Actions** > **Encrypt**
4. Select the check box next to the device you want to encrypt.
5. Click **Encrypt**.
6. Repeat for each SSD requiring encryption.

Encrypt single devices using the CLI

Use this procedure to encrypt single devices using the CLI:

1. Connect to the SDS using SSH.
2. To encrypt the new drive, run:

   ```
   svm encrypt <device_name>
   ```

   where *<device_name>* is a variable.

   For example:

   ```
   svm encrypt /dev/sdX
   ```

   or

   ```
   svm encrypt /dev/nvmeXn0
   ```

   After about 60 seconds, the new device will be encrypted.

Encrypt multiple devices using the CLI

Use this procedure to encrypt multiple devices using the CLI.

1. Connect to the SDS node using SSH.
2. Encrypt the drives.

- SAS-based SSDs:
  a. Select the devices for encryption.

     For example, `/dev/sdX`, where *X* could be a b c - z.

  b. Run the following command:

     ```
     for i in b c {j..o} {r..v};do svm -y encrypt /dev/sd$i;done
     ```
- NVMe-based SSDs:
  a. Select the devices for encryption.

     For example, `/dev/nvmeXn1`, where *X* could be 0 1 2 - 24.

  b. Run the following command:

     ```
     for i in {0..3} 6 7 {10..13} 16 17;do svm -y encrypt /dev/nvme$[i]n1;done
     ```

After about 60 seconds, the new device will be encrypted.

Verify SED encryption

Find the new device path and confirm that the SED is encrypted.

1. In CloudLink Center, click **Agents** > **Machines**.
2. Select the host of the encrypted device.
3. Use the device name listed in the `svm status` command output to perform the following verification checks:
   - In the **Devices** area, verify that the drive is visible and has a status of **Encrypted HW**.
   - In the **SED** area, verify that the status of the drive is displayed as **Managed**.

Manage self-encrypting storage devices

CAUTION: Never attempt to manage or erase the devices that are currently attached to the SDS. Devices should only be managed before they are added to the SDS, or after removing them from the SDS. Any existing data on the device will be destroyed as a result of this procedure.

- Manage the SEDs (self-encrypting devices) using one of the following methods:

Manage the SEDs in CloudLink Center

Manage the device from CloudLink Center.

1. Click **Agent** > **Machines**.
2. Select the check box next to the SED you want to manage.
3. Click **Actions** > **Manage SED**.
   The **Manage SED** dialog box is displayed. It is populated with the first unmanaged SED device. You can add different device to encrypt.
4. Click **Manage**.
5. Repeat for each SED requiring encryption.

Manage single SEDs using the CLI

Manage individual devices using the CLI:

1. Connect to the SDS node using SSH.
2. Run the following command to discover the unencrypted and unmanaged SEDs:

   ```
   svm status
   ```

3. Run the following command so that CloudLink will control the SED device:

```
svm manage /dev/sdX
```

For example:

```
svm manage /dev/sdh
```

4. Run `svm status` again to verify that the device is now managed:

The output should be similar to the following:

```
[root@rr5-o25-r650-node-03 ~]# svm status
State: Connected (server 100.65.23.225)
Group: Default
Policy: Manual
AES-NI HW acceleration: Yes

Volumes:
/                   unencrypted
swap                unencrypted

Devices:
/dev/sdf            managed             (sed    SZ: 1788G MOD: KPM6WRUG1T92     SPT:
Yes  )
/dev/sdd            managed             (sed    SZ: 1788G MOD: KPM6WRUG1T92     SPT:
Yes  )
/dev/sdb            managed             (sed    SZ: 3577G MOD: KPM6WVUG3T84     SPT:
Yes  )
/dev/sde            managed             (sed    SZ: 1788G MOD: KPM6WRUG1T92     SPT:
Yes  )
/dev/sdc            managed             (sed    SZ: 3577G MOD: KPM6WVUG3T84     SPT:
Yes  )
/dev/sdb            unencrypted      (sds    SN:94917674              )
/dev/sdc            unencrypted      (sds    SN:94917675              )
/dev/sdd            unencrypted      (sds    SN:94917676              )
/dev/sde            unencrypted      (sds    SN:94917677              )
/dev/sdf            unencrypted      (sds    SN:94917678              )
/dev/sdg            encrypted        (sds    SN:94917679             /dev/mapper/
svm_sdg)
/dev/sdh            encrypted        (sds    SN:94917680             /dev/mapper/
svm_sdh)
/dev/sdi            encrypted        (sds    SN:94917681             /dev/mapper/
svm_sdi)
/dev/sdj            encrypted        (sds    SN:94917682             /dev/mapper/
svm_sdj)
/dev/sdk            encrypted        (sds    SN:94917683             /dev/mapper/
svm_sdk)
```

NOTE: The status of the SED devices is displayed in the output as managed, but unencrypted.

Verify SSD or NVMe encryption

Find the new device path of the SSD or NVMe and confirm that the device is encrypted.

1. To find the new device path, run:

```
svm status
```

Output similar to the following should be displayed:

- SSD: `/dev/mapper/svm_sdh`
- NVMe: `/dev/mapper/svm_nvme6n1`

2. In CloudLink Center, click **Agents** > **Machines**.
3. n the **Devices** area, verify that the drive is visible and has a status of **Encrypted**.

Add the encrypted devices to an SDS

After the devices are encrypted, in PowerFlex Manager add them to the relevant PowerFlex SDS. For software-encrypted SSDs, use their new `/dev/mapper/..` path when adding the device. SEDs use the normal `/dev/sdX` path.

NOTE: Do not add a software-encrypted SSD using its original path (`/dev/...`). Doing so can stop the device encryption (device erasure, in terms of CloudLink), and this can corrupt the data on the device.

Remove software-encrypted devices from an SDS and remove device encryption

Use this task to remove a device from an SDS and remove the device's encryption. Use this task for both software-encrypted devices and SEDs.

CAUTION: Never attempt to manage or erase the devices that are currently attached to the SDS. Devices should only be managed before they are added to the SDS, or after removing them from the SDS. Any existing data on the device will be destroyed as a result of this procedure.

1. Remove the device from the SDS:
   - PowerFlex CLI:

     Use the SCLI `--remove_sds_command`. For example:

     ```
     scli --remove_sds_device --sds_name MY_SDS --device_path /dev/mapper/svm_sdX
     ```

     or

     ```
     scli --remove_sds_device --sds_name MY_SDS --device_path /dev/mapper/svm_nvmeXn1
     ```
2. Remove the encryption through CloudLink Center GUI or using the svm `erase` command:

   NOTE: Removing (erasing) a device from CloudLink destroys all data on the device.

   ```
   svm erase <device_name>
   ```

   For example:

   ```
   svm erase /dev/sdX
   ```

   or

   ```
   svm erase /dev/nvmeXn0
   ```
3. Confirm that the device was erased.

8

Modifying DNS entries

This section contains procedures on how to change the DNS entries. If you have multiple DNS servers that are configured but want to use a specific server, you can change the DNS entries to point only to that preferred server instead of listing all available servers.

Modifying the DNS entries in the embedded Linux operating system

Modify or update the DNS entries in the embedded Linux operating system.

1. SSH to the embedded Linux operating system running on the SDS.
2. Go to the file path: `sudo vi /etc/sysconfig/network/config`
3. Add or modify the line:
   - **`NETCONFIG_DNS_STATIC_SEARCHLIST=xxx.com`**, where *xxx.com* is the domain name
   - **`NETCONFIG_DNS_STATIC_SERVERS=x.x.x.x y.y.y.y`**, where *x.x.x.x y.y.y.y* is the DNS server IP address
4. Press **Esc** and type **`:wq`** to save the file.
5. Type **`sudo netconfig update -f`** to refresh and update the configuration.

Modifying DNS entries in the management virtual machines

Modify or update the DNS entries in the management virtual machines.

1. Establish an SSH connection to each of the management virtual machines nodes.
2. To view the existing DNS entries, run:

   ```
   sudo cat /etc/sysconfig/network/config | grep -i NETCONFIG_DNS_STATIC
   ```

3. To add or update the DNS entries:
   a. Run the following command:

      ```
      sudo vi /etc/sysconfig/network/config
      ```

   b. Add or modify the following lines:
      - **`NETCONFIG_DNS_STATIC_SEARCHLIST=example.com`**, where *example.com* is the domain name
      - **`NETCONFIG_DNS_STATIC_SERVERS=x.x.x.x y.y.y.y`**, where *x.x.x.x y.y.y.y* are the DNS server IP addresses.

   c. Press **Esc** and type **`:wq`** to save the file.
   d. To refresh and update the configuration, run:

      ```
      sudo netconfig update -f
      ```

4. Update the **coredns** pods on all the management virtual machines after making the DNS entry change for the pods in internal resolutions. Run the following command to view the pod names:

   ```
   kubectl get pods -n kube-system | grep rke2-coredns-rke2-coredns
   ```

5. To restart the **coredns** pods, delete them by running the following command:

```
kubectl delete pods <pod-name> -n kube-system
```

The pods are recreated after they are deleted.

Modifying the DNS entries in VMware vCenter

Modify the VMware vCenter DNS using VAMI.

1. Log in to VMware vCenter VAMI: **`https://vcenter_ip_or_FQFN:5480`**.
2. Go to the **Networking** section and in **Network Settings**, click **Edit**.
3. From **Edit Network Settings**, select **Network Adapter** and click **Next**.
4. In **Edit Settings**, expand the **Hostname and DNS** field, make the changes, and click **Next**.
5. In **SSO credentials**, enter the administrator credentials, for example, **`administrator@vsphere.local`**.
6. In **Ready to Complete**, review the changes, accept the acknowledgment, and click **Finish**.
7. A window appears with the message **Network update in progress**. When complete, the window displays **Network update succeeded**, click **Close**.

Modifying the DNS entries in the VMware ESXi host

1. Log in to the VMware ESXi host: **`https://ESXi_host_ip`**.
2. Click **Networking** > **TCP/IP Stacks**. Select **Default TCP/IP stack** and click **Edit Settings**.
3. Under **Basic DNS Configuration**, select **Manually configure the settings for this TCP/IP stack**.
4. Add or update the DNS IP addresses in the **Primary DNS server** and **Secondary DNS server** fields. In the **Search domain** field, add the domain name and click **Save**.

9

Backing up and restoring

Use the following procedures to back up and restore PowerFlex Manager, PowerFlex components, and CloudLink Center. Backups are critically important because they enable you to restore to a previously known state.

Back up and restore

Creating a backup saves all user-created data to a remote share from which it can be restored. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Perform frequent backups to guard against data loss and corruption. The best practice is to take a snapshot of PowerFlex Manager every time you perform a restore.

The **Backup** page displays information about the last backup operation that was performed on PowerFlex Manager. The information provided applies to both manual and automatically scheduled backups and includes the following:

- Last backup date
- Last backup status
- Back up directory path to a CIFS share
- Back up directory username

The **Backup** page also displays information about the status of automatically scheduled backups (enabled or disabled).

On this page, you can:

- Manually start an immediate backup
- Edit general backup settings
- Edit automatically scheduled backup settings

After performing a backup operation, you need to run a script outside of PowerFlex Manager to restore from the backup. The user interface does not support the ability to restore from a backup.

Editing backup settings and details

You can change the location where backup files are saved or the password that is required to access a backup file.

1. On the menu bar, click **Settings** and then click **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Settings**.
4. Indicate the network share location where the backup file is saved, enter a backup directory path in the **Backup Directory Path** box.

   Use the following format:

   - CIFS—`\\host\share`
5. If the username and password are required to access the network share, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes.
6. Click **Test Connection** to confirm that the backup settings you provided are correct.
7. To open the backup file, enter a password in the **Encryption Password** box.
8. To verify the encryption password, enter the password in the **Confirm Encryption Password** box.
9. To schedule automatic backups, next to **Scheduled Backups**, select **Enabled**. To discontinue automatically scheduled backups, deselect **Enabled**.
10. To specify the days on which backup must occur, select the days under **Frequency**.
11. From the **Run Time** drop-down list, select the time.
12. Click **Save**.

Backing up

In addition to automatically scheduled backups, you can manually run an immediate backup.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Now**.
4. Select one of the following options:
   - To use the general settings that are applied to all backup files, select **Use Backup Directory Path and Encryption Password from Settings and Details**.
   - To use custom settings:
     a. In the **Backup Directory Path** box, enter a file path where the backup file is saved. Use this format:
        - CIFS—`\\host\share`

     b. Optionally, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes, if they are required to access the location you provided.
     c. Click **Test Connection** to confirm that the backup settings you provided are correct.
     d. In the **Encryption Password** box, enter a password that is required to open the backup file, and verify the encryption password by entering the password in the **Confirm Encryption Password** box.
5. Click **Backup Now**.

Restoring PowerFlex Manager

Restoring PowerFlex Manager returns user-created data to an earlier configuration that is saved in a backup file. To restore from a backup, you must run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Before you begin the restore procedure, you must satisfy these prerequisites:

- The restore cluster must be the same PowerFlex version and Kubernetes version.
- The restore cluster must have the same IP addresses and configuration.

  The cluster configuration must be the same as the cluster configuration where the backup was taken.

  - All Kubernetes nodes must have the same IP addresses.
  - All Kubernetes nodes must have the same names.
  - All LoadBalancer IP addresses must be the same.

CAUTION: Restoring an earlier configuration restarts PowerFlex Manager and deletes data that is created after the backup file to which you are restoring. Any running jobs could be terminated as well.

1. Log in, using the username and password used for the installer VM, to the node where the PowerFlex Manager platform installer was initially run.
2. Run the restore script that is included with the installer bundle:

   `./restore_backup.sh`
3. Provide details to complete the execution of the script.

   You are prompted to provide the SSH username and password. In addition, you are asked to specify whether the passwords are the same for all nodes, and also provide the location of the backup file and the encrypted password for the file. You may also be asked to provide the CIFS username and password. The CIFS credentials may not be required for a CIFS share that allows for anonymous access.

   NOTE: The passwords must be in base64 encoded format.

   To complete the execution of the restore script, you must specify whether the restore operation will be performed on an existing cluster or a new cluster.

   Here is a snippet that shows a sample that is run of the restore script:

```
[root@sheetal-installer scripts]# ./restore_backup.sh

/usr/local/lib/python3.8/site-packages/paramiko/transport.py:236:
CryptographyDeprecationWarning: Blowfish has been deprecated "class":
```

```
algorithms.Blowfish, Installation logs are available at <Bundle root>/PFMP_Installer/
logs/ directory. More detailed logs are available at <Bundle root>/atlantic/logs/
directory. PFMP Installer is about to reset a PFMP cluster based on the configuration
specified in the PFMP_Config.json.

Please enter the ssh username for the nodes specified in the
PFMP_Config.json[root]:root

Are passwords the same for all the cluster nodes[Y]?:Y

Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Password:

Please enter backup file location - Example CIFS: \\1.2.3.4\Lab\backup\file.tgz

Please enter CIFS username. Press enter to skip if username is not
required:administrator

Please enter CIFS password(base64 encoded). Press enter to skip if username is not
required: UmFpZDR1cyE=

Please enter encryption password for backup file (base64 encoded): UmFpZDR1cyE=

Perform restore on existing cluster? Please enter yes/y or no/n :
```

The restore process prints status information until the restore is complete.

Related tasks

Restoring compliance file data

Restoring compliance file data

Use this procedure to return compliance files to a healthy state.

After performing a restore operation, the compliance files will be in an ERROR state. The compliance file data is not part of the backup payload, so after a restore, the compliance files appears in an error/unhealthy state. Any changes previously made to compliance files using single component upgrade, for example BIOS, iDRAC, or ESXi version modifications, will not be preserved through the backup and restore process.

1. Resynchronize the compliance file from the original source to restore it to a good state.
2. If any components were previously modified using single component upgrade (BIOS, iDRAC, ESXi depot), edit the compliance file using single component upgrade to reapply those changes.
3. Confirm that the compliance file is no longer in ERROR state and all components reflect the expected versions.

Related tasks

Restoring PowerFlex Manager

Back up and restore using VM snapshots

If the PowerFlex management platform is running on VMs, you can back up and restore using the VMware VM snapshots once you shut down the database and VMs.

If the cluster is deployed on VMware VMs, you can take the snapshots from the vSphere Client user interface, or you can use a CLI command on the ESXi system that is hosting the VMs.

For VM snapshots, the database and VMs must be shut down first before a reliable snapshot can be taken.

CAUTION: When you shut down the database and VMs, you shut down the PowerFlex management platform. The PowerFlex Manager user interface is not accessible and any running jobs fail when the system comes back on. Only after the database is shut down, should the VMs be shut down or the snapshots taken.

Follow the steps below to shut down the database and VMs.

1. To set the default namespace, enter:

```
# export NAMESPACE=powerflex
```

2. To set the Database Smart Assistant (DBSA) IP address variable, enter:

```
# export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o
jsonpath='{.spec.clusterIP}')
```

3. To verify that the database cluster is up and running, enter:

```
# curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
```

4. To gracefully shut down the database cluster, enter:

```
# curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
```

5. To take the snapshots and adjust the resources, shut down the VMs.
6. Power the VMs back on.
7. To set the default namespace, enter:

```
# export NAMESPACE=powerflex
```

8. To set the Database Smart Assistant (DBSA) IP address variable, enter:

```
# export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o
jsonpath='{.spec.clusterIP}')
```

9. To start the cluster back up, enter:

```
# curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"
```

10. To verify that the cluster startup occurred, enter:

```
# curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
```

To restore to the previous point in time, you can revert the VMs from the snapshots in the VMware vSphere Client user interface.

Integrated Dell Remote Access Controller 9 (iDRAC9)

You can back up and restore iDRAC configuration settings as needed.

Back up iDRAC

You can back up the system and configuration settings for iDRAC. The backup creates a file that you can save locally or on a network share (CIFS, NFS, HTTP, or HTTPS).

1. Log in to the iDRAC using root credentials.
2. Click **Configuration** > **Server Configuration Profile**.
3. Click **Export**.
4. Select one of the following to specify the location type:

   CIFS, NFS, HTTP, or HTTPS

   - **Local** to save the configuration file on a local drive.
   - **Network Share** to save the configuration file on a CIFS or NFS share.
   - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If the proxy is configured for HTTP/HTTPS, proxy settings are also required.

5. Select the components whose configuration you must back up.
6. Select the **Export** type:
   - **Basic**
   - **Replacement Export**
   - **Clone Export**
7. Select an **Export file format**.
8. Select **Additional export items**.

Restore iDRAC

You can restore iDRAC from a backup when you want to use previous configuration settings.

1. Click **Configuration** > **Server Configuration Profile**.
2. Select one of the following to specify the location type:
   - **Local** to save the configuration file on a local drive.
   - **Network Share** to save the configuration file on a CIFS or NFS share.
   - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

   Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If the proxy is configured for HTTP/HTTPS, proxy settings are also required.
3. Select the components listed in the **Imported Components** option.
4. Select the **Shutdown** type.
5. Select the maximum wait time to specify the wait time before the system shuts down after the import is complete.
6. Click **Import**.

Network switch configuration back up and restore

Back up and restore running configuration of the Cisco and Dell switches.

If you own the switches, take the switch backup immediately after successful PowerFlex deployment.

NOTE: You are responsible for providing the backup location and maintaining the backup as well. Dell Technologies recommends storing the backup in a separate shared location, for example, not on the jump server.

1. Connect to the Cisco Nexus or Dell PowerSwitch switch, either using the console cable, Telnet, or SSH using admin credentials, type:

   ```
   #copy running-config scheme://server/[url/]filename
   ```

   For the scheme argument, you can enter **tftp:**,**ftp:**, **scp:**, or **sftp:**.

   The server argument is the address or name of the remote server, and the URL argument is the path to the source file on the remote server. The server, URL, and file name arguments are case-sensitive.

   For example:

   ```
   switch# copy running-config tftp://10.10.10.1/sw1-run-config.bak

   switch# copy running-configuration scp://root:calvin@10.11.10.12/tmp/backup.txt
   ```

2. Restore the network configuration, connect to the Cisco Nexus or Dell PowerSwitch switch, either using the console cable, Telnet, or SSH using admin credentials, type:

   ```
   #copy scheme://server/[url/]filename running- config
   ```

   For the scheme argument, you can enter **tftp:**,**ftp:**, **scp:**, or **sftp:**.

   The server argument is the address or name of the remote server, and the URL argument is the path to the source file on the remote server. The server, URL, and file name arguments are case-sensitive.

   For example:

```
switch# copy tftp://10.10.10.1/my-config running-config
switch# copy scp://root:calvin@10.11.10.12/tmp/backup.txt running-configuration
```

Backing up and restoring CloudLink Center

Use this procedure to create a CloudLink Center backup file.

View backup information

Viewing backup information helps ensure data recovery, supports compliance, aids troubleshooting, enables version control, and facilitates system upgrades or migrations.

To view the backup information, log in to the CloudLink Center, and click **System** > **Backup**. The **Backup** page lists the following information:

- **Backup File Prefix**—Prefix used for the backup files.
- **Current Key ID**—The identifier for the current RSA-2048 key pair.
- **Current Backup File**—The name of the current backup file.
- **Current Backup Time**—The date and time that the current backup file was generated.
- **Backup Schedule**—The schedule for generating automatic backups.
- **Next Backup In**—The time remaining before the next automatic backup is generated.

When a backup file is downloaded, the **Backup** page lists the following additional information:

- **Last Downloaded File** —The name of the backup file that was last downloaded. Only shown when a backup file has been downloaded.
- **Last Downloaded Time** —The date and time of the last backup file download. Only shown when a backup file has been downloaded.
- **Backup Store** —The backup store configuration type. If you have not configured a backup store, the value is Local, which is stored on the local desktop.

  You can also use the FTP or SFTP servers as backup stores. To change the backup store, click **Actions** > **Change Backup Store**.

  If you have configured an FTP or SFTP backup store, the following additional information is available:

  - **Host** —The remote FTP, SFTP, or FTPS host where you saved the CloudLink Center backups. You can set this value to the host IP address or hostname (if DNS is configured).
  - **Port**—The port used to access the backup store.
  - **User**—The user with permission to access the backup store.
  - **Directory**—The directory in the backup store where backup files are available.

Change the schedule for automatic backups

CloudLink Center automatically generates a backup file each day at midnight (UTC time).

To change the schedule for generating automatic backups, click **System** > **Backup** > **Actions** > **Change Backup Schedule**.

Generate a backup file manually

Use this procedure to generate a backup manually, if you want to preserve CloudLink Center before the next automatic back up. Download the backup file when you generate a backup file manually.

1. Log in to the CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Generate New Backup**. A backup file is generated.

Generate a backup key pair

You can generate a new backup key pair. For example, if the private key for a backup key pair is lost, you can generate a new key pair. You cannot access your backup files without the associated private key. When you generate a new key pair, CloudLink Center automatically generates a new backup file to ensure that the current backup can be opened with the private key of the current key pair.

Dell Technologies recommends the following practices when you generate a new backup key pair.

1. Download the private key to the Downloads folder for the current user account. For example, `C:\Users\Admnistrator\Downloads`.

   NOTE: The previously generated backup key will not open the backup file created, after a new key is generated.

2. Generate the new key pair by clicking **System** > **Backup** > **Actions** > **Generate and Download New Key**.
3. Click **Generate and Download**.

Download the current backup file

You can download the current backup file at any time. The current backup file is either:

- The last backup file that CloudLink Center automatically created.
- The last backup file that you manually generated after the last automatic backup.

To download the backup file:

1. Click **System** > **Backup** > **Actions** > **Download Backup**.
2. In the **Download Current Backup** dialog box, click **Download**.

When you download the current backup file, CloudLink Center shows the age of the backup file.

Restore the CloudLink backup

Use this procedure to restore the CloudLink backup.

1. Log in to the CloudLink Center.
2. Click **System** > **Backup** > **Actions** > **Restore Keystores**.
3. In the **Restore Keystores** dialog box, complete the following steps:
   a. In the **Key** box, browse to the private key file.
   b. In the **Backup** box, browse to the backup file.
   c. In the **Unlock** box, type the passcode that was set during the initial configuration of the CloudLink Center.
   d. Click **Restore**.
   A **Restore Keystores succeeded** message is displayed.

   NOTE: If the CloudLink backup is not associated with a key pair, the file is corrupted or key mismatch error message is displayed. In such a scenario, generate a new key pair and download the backup file.

10

Managing usernames and passwords

This section contains procedures to change iDRAC, VMware vSphere Client, CloudLink and system passwords, generate SSH key pairs, and manage credentials.

PowerFlex servers

Change the Integrated Dell Remote Access Controller 9 (iDRAC9) password

Use the iDRAC web console to change the root password.

1. Open a web browser and type: **`http://<ip_address_of_iDRAC>`** .
2. Log in as **`root`**.
3. Expand **iDRAC Settings**.
4. Click **Users** > **Local Users**.
5. Select **User ID 2** and click **Edit**.

   ID 2 is the root user.

6. Under **User Configuration**, in the **User Account Settings** section, change the password.
   a. In the **Password** box, enter the new password.
   b. In the **Confirm Password** box, re-enter the new password.
7. Click **Save**.

Change the system and setup passwords

Use this procedure to change the system BIOS password.

If the password status is set to `locked`, you cannot change or delete the system or setup password.

1. Enter System Setup by pressing **F2** immediately after turning on or restarting your system.
2. On the System Setup Main Menu, click **System BIOS** > **System Security**.
3. On the **System Security** screen, ensure that **Password Status** is set to **`Unlocked`**.
4. In the **System Password** field, change or delete the existing system password and press **Enter**.
5. In the **Setup Password** field, change or delete the existing setup password and press **Enter**.

   If you change the system or setup password, a message prompts you to re-enter the password. If you delete a password, a message prompts you to confirm the deletion.

6. Press **Esc** to return to the **System BIOS** screen. Press **Esc** again. A message prompts you to save the changes.

Generate SSH key pairs (Optional)

To create the SSH keys for a PowerFlex node, switch **OS Admin** and **OS User**. You can manually generate the SSH keys that are of type RSA and ensure that the key is generated without a pass phrase.

1. To generate a private key, run:

   `ssh-keygen -b 2048 -t rsa`

2. Enter a file path to save the public and private keys or proceed with the default file path without a pass phrase:

Sample output:

```
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase: # do not enter a passphrase
Enter same passphrase again: # do not enter a passphrase
```

Sample output that indicates the public and private keys are saved in the default file path:

```
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:fKIllqp5UXeuEW7xuYeE7PNHpvjAUatNx6I/bisRqRU root@PFMP2-4
The key's randomart image is:
+---[RSA 2048]----+
|                 |
|        E        |
|         o.      |
|      .oB..o     |
|     .+*SO=.o    |
|    .o.=BO=oo    |
|    ...o=*.*     |
|   o.   *o* o    |
|  o.     B*=     |
+----[SHA256]-----+
```

3. Verify that the public and private keys are saved in the default file path: `/root/.ssh/`.
   Sample output:

```
cd /root/.ssh
ls
```

- `id_rsa.pub` is an indication of a public key.
- `id_rsa` is an indication of a private key.

Virtualization

Changing a VMware ESXi host root password

For security reasons it might be necessary to change the password for the root user on a VMware ESXi host after installation.

Use any of the following methods to change the root password for the VMware ESXi host:

- VMware vSphere Client
- VMware ESXi shell command
- VMware ESXi host System Customization menu

Change the password using VMware vSphere Client

1. Log in to the host UI https://esxi_IP_address using root credentials.
2. In the left pane, under **Host**, click **Manage**.
3. Select the **Security & Users** tab, click **Users**, and click the **root** user.
4. Click **Edit User**.
5. In the **Edit User** dialog box, enter and confirm a new password.
6. Click **Save**.

Change the password using the VMware ESXi shell command

Log in to the VMware ESXi host service console as **root user**.

1. When prompted, enter the current password.
2. To change the root password, enter: **passwd root**.
3. Enter the new root password and press **Enter**.
4. Verify the password by entering it again.

Change the password using the VMware ESXi host system customization menu

Log in to the VMware ESXi host service console as **root** user.

You can also acquire root privileges by running the **su** command.

1. From the **System Customization** menu of the VMware ESXi host, use the keyboard arrows to select **Configure Password** and press **Enter**.
2. In the **Configure Password** dialog box, fill in the required fields to change the password:
   a. Enter the **Old Password** of the VMware ESXi host.
   b. Enter the new root password in the **New Password** field. Re-type it in the **Confirm Password** field.
   c. Press **Enter**.

Modify the VMware vCenter Server single sign on default administrator password

Use this procedure to change the default password for the VMware vCenter single sign on administrator account.

Log in to the VMware vSphere Client and connect to vCenter.

1. In the left pane, select **Administration**.
2. Under Single Sign On, select **Users and Groups**.
   The **admin** user displays in the right pane.
3. On the **Users** tab, select the domain from the drop-down menu, and select the **administrator** user. Click **Edit**.
4. Set and confirm the password for the admin user account. Be sure to use a strong password as the system validates the password before accepting it.
5. Click **Save**.

Change the administrator password for VMware vCenter Server Appliance

Use this procedure to change the VMware vCenter Server Appliance administrator password.

1. Log in to the VMware vCenter Server Appliance Web console.
2. On the **Admin** tab, enter your current password in the **Current administrator password** box.
3. Enter the new password in the **New administrator password** and **Retype new administrator password** boxes.
4. Click **Change password**.

Resetting the VMware vCenter Server Appliance root password

Use this procedure to reset a forgotten vCenter Server Appliance root password.

1. Take a snapshot or backup of the vCenter Server Appliance. Do not skip this step.
2. Reboot the vCenter Server Appliance.
3. After the operating system starts, press the **E** key to access the **GNU GRUB Edit Menu**.
4. Locate the line that starts with the word `Linux`. Append the following entries to the end of the line:

   ```
   rw init=/bin/bash
   ```

5. Press **F10**.

6. At the command prompt, type the following command: **passwd**
7. Enter a new root password and re-enter the password to confirm it.
8. Unmount the file system by running the following command: **umount /**
9. Reboot the vCenter Server Appliance by running the following command: **reboot -f**
10. Confirm that you can access the vCenter Server Appliance using the new root password.
11. Remove the snapshot taken in step 1 if applicable.

Management VMs

Use this procedure to change users passwords for the management VMs.

The default username is delladmin.

1. Log in to the appropriate user account. If changing a password for another user, log in as root.
2. Open a shell prompt and type one of the following commands:
   - To change your own password: passwd
   - To change the password of another user: passwd username

   When prompted, enter and confirm the new password.

Change CloudLink passwords

CloudLink software is installed on the CloudLink Center VMs.

Unlock secadmin user password

Use this procedure to unlock the secadmin password.

1. Log in to the controller vCenter web client, and launch the CloudLink Center VM console.

   NOTE: For a single vCenter environment, log in to single vCenter with a controller and customer datacenter. Go to the controller datacenter to launch the CloudLink center VM console.

2. Log in using the CloudLink user credentials.
3. In the **Update Menu** dialog box, select **Unlock User**, and click **OK**.
   A user unlocked message is displayed.
4. Click **OK**.

Changing the operating system password

If you lose the password on storage VM (SVM), start up in recovery mode to reset it.

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **SDS**.
3. Select **Node** > **More actions** > **Enter Maintenance Mode**.
4. Log in to VMware vCenter. Select the cluster and click **Related Objects** > **Virtual Machines** > **Open Console**.
5. Restart the SVM.
6. Enter **ESC** during restart and click **OK** to enter text mode.

   NOTE: You might have to hold down the **ESC** key.

7. Type `E` to edit the password.
8. Select the second line (starts with `kernel /boot…`) and type `E`.
9. Append `init=/bin/bash` to the end of the line.
10. Select `B` to restart in recovery mode.
11. After the PowerFlex node restarts, type `passwd`, and enter a new password.

12. Type `reboot`, and allow the SVM to restart using the new password.

11

Retrieving logs

This section describes how to retrieve logs for an operating system, components, and PowerFlex software.

Retrieving logs for an operating system

This section contains instructions to retrieve logs from a VMware ESXI-based operating system, a Linux-based operating systems, and a Windows-based operating system. It also contains a procedure on how to configure and retrieve operating system crash dumps.

Retrieve logs from a VMware ESXi-based operating system

Use this task to retrieve the logs from a VMware ESXi-based PowerFlex node.

Ensure that you have access to the following:

- IP address of the node
- Password for the root user

When the operating system is started, the ESXi operating system logs are collected in the system-defined folders `/var/log` and `/scratch/log/` on the local drive.

1. Log in to the PowerFlex node.
2. From ESXi, run the `scp` command to collect the ESXi operating system logs from the following folders on the local drive:
   - `/var/log`
   - `/scratch/log/`

   For additional data, see KB article: Collecting diagnostic information for VMware ESXi using vSphere Client

An inventory job is scheduled to run every day at [HH:mm] to identify the resources that are running in your system. If an upgrade operation is in progress simultaneously, the upgrade operation overrides the job and the job is skipped.

If you turn off the inventory job setting, the job is paused for 24 hours. After 24 hours, the inventory job setting automatically turns back on and the inventory job continues running from the time it was paused.

Retrieve logs from a Linux-based operating system

Use this task to retrieve the logs from a Linux-based PowerFlex node.

Ensure that you have access to the following:

- IP address of the node
- Password for the root user
- Grab Utility, downloadable from the Dell Technologies Support site

When the Linux server is started, the Linux logs are collected in the system-defined directory `/var/log`, on the local drive. Operating system logs are also collected automatically when running the get_info log collection process.

For information on running get_info, see Collecting debug information using get_info.

1. Log in to the PowerFlex node.
2. Run the `scp` command to collect the Linux operating system logs from the `/var/log` directory on the local drive.

   The `/var/log` folder provides logs for preliminary system investigation.
3. To retrieve full logs, including customer- and OS-related information, run the Grab utility.

Retrieve logs from Windows-based operating system

Use this task to retrieve the logs from a Windows-based PowerFlex node.

Ensure that you have access to the following:

- IP address of the node
- Password for the administrator user
- Grab utility available on the Dell Technologies Support site.

1. Log in to the PowerFlex node.
2. Run the Windows Event Viewer.

   The **Event Viewer** screen displays **Event Viewer (Local)** as the root node.
3. In the navigation pane, go to the **Windows Logs** node and select the **System** sub node.
   The **Actions** pane is displayed on the right side of the screen.
4. From the **Actions** pane, click **Save All Events As**.
   The navigation tree with your local folders is displayed.
5. Browse to the intended location on your local drive, give the file an appropriate name, and select the file format.
6. Click **Save**.
   The Windows OS-related log files are saved on your local drive.

   The file on the local drive provides logs for preliminary system investigation.
7. To retrieve full logs, including customer and OS-related information, run Grab utility.

Configure and retrieve operating system crash dumps

Use this task to configure and retrieve the operating system crash dumps.

Ensure that you have access to a web browser.

1. Open your web browser.
2. To configure and retrieve the crash dumps, click an appropriate link that matches with your operating system:

Operating System	Link for crash dump configuration procedure
ESXi	Configuring an ESXi/ESX host to capture a VMkernel coredump
Linux (RedHat)	How to troubleshoot kernel crashes, hangs, or reboots with kdump on Red Hat Enterprise Linux
Linux (SLES)	Configure crashkernel memory for kernel core dump analysis
Windows	Generate a kernel or complete crash dump

3. To analyze the crash dumps, click an appropriate link that matches with your operating system:

Operating System	Link for crash dump configuration procedure
Linux (RedHat)	How can I use crash to send Red Hat some vmcore pre-analysis information before or while uploading the vmcore image?
Linux (SLES)	Generating a Kernel Core Dump Analysis File
Windows	How to read the small memory dump file that is created by Windows if a crash occurs

Retrieving logs for components

This section contains instructions to retrieve logs for components. You can retrieve the log files for components manually or automatically. You can collect the logs of one component at a time or of all system components at one time. You can also enable automatic log collection that collects logs automatically when triggered by system alerts.

Retrieve the PowerFlex core component logs

You can retrieve the PowerFlex component logs manually, directly from the component server, one node at a time.

Ensure that you have the following user login credentials:

- IP address of the node
- Password for the root user in ESXi- and Linux-based systems and the administrator user in Windows-based systems

The PowerFlex logs relate to specific components (MDM, SDS, SDR, SDT, or LIA) on each server node and its operating system environment.

1. SSH or RDP to the PowerFlex node.
   In ESXi-based PowerFlex, log in to the Linux-based SVM.
2. Run the script for the PowerFlex component.

Operating System	Run this script
ESXi or Linux	`/opt/emc/scaleio/<PowerFlex component>/diag/get_info.sh` where the value of the PowerFlex component is mdm, sds, sdr, sdt, or lia
Windows	`"C:\Program Files\EMC\scaleio\sdc\diag\get_info.bat" -f`

   The get_info syntax is explained fully in Collecting debug information using get_info.

   - If the selected node is the primary MDM, use the flags `-u <MDMuser> -p <MDMpassword>`, instead of `-f`.
   - If the selected node contains more than one PowerFlex component, running any script will gather logs for all components on that node.

   When the log collection process is complete, an archive file (either TGZ or ZIP) containing the logs of all PowerFlex components in the node, is created in a temporary directory. By default, the directory is `/tmp/scaleio-getinfo` on Linux hosts or `C:\Windows\Temp\ScaleIO-getinfo` on Windows hosts.
3. Verify that output similar to the following is returned, which shows that the process of log collection was completed successfully:
   bundle available at `'/tmp/scaleio-getinfo/getInfoDump.tgz'`
   NOTE: The script can generate numerous lines of output. Therefore, look for this particular line in the output.
4. Retrieve the log file.

Related information

Collecting debug information using get_info

Collecting debug information using get_info

The get_info script is a debug utility which allows a technician to collect debug information for customer support analysis.

You can run the script once at a given time. If there is insufficient drive space then the script will not run.

Syntax

```
get_info.sh [OPTIONS]
```

Optional parameters:

-a, --all
Collect all data

-A, --analyse-diag-coll
Analyze diagnostic data collector (diag coll) data

-b[COMPONENTS], --collect-cores[=COMPONENTS]

Collect existing core dumps of the space-separated list of user-land components, COMPONENTS (default: all user-land components)

For example, `-b'mdm sds'` (no space between option name and COMPONENTS)

For example, `--collect-cores='mdm sds'` (separate option name and COMPONENTS with a single equal sign, "=")

-d OUT_DIR, --output-dir=OUT_DIR

Store collected bundle under directory OUT_DIR (default: <*WORK_DIR*>/scaleio-getinfo, see `--work-dir` for the value of <*WORK_DIR*>)

-f, --skip-mdm-login

Skip query of PowerFlex login credentials

The parameters `-k NUM` and `--max-cores=NUM` collects up to NUM core files from each component (default: all core files)

Implies --collect-cores -l, --light

Generate light bundle (not recommended)

--ldap-authentication

Log in to PowerFlex using LDAP-based authentication

-m NUM, --max-traces=NUM

Collect up to NUM PowerFlex trace file from each component (default: all files)

--management-system-ip=ADDRESS

Connect to SSO or management at `ADDRESS` for PowerFlex login (default: scli default)

--mdm-port=PORT

Connect to MDM using PORT for SCLI commands (default: scli default)

-n, --use-nonsecure-communication

Connect to the MDM in non-secure mode

-N, --skip-space-check

Skip free space verification

--overwrite-output-file

Overwrite output file if it already exists

-p PASSWORD, --password=PASSWORD

Use PASSWORD for PowerFlex login (default: scli default)

--p12-password=PASSWORD

Encrypt PowerFlex login PKCS#12 file using PASSWORD (default: scli default)

--p12-path=FILE

Store PowerFlex login PKCS#12 file as FILE (default: scli default)

-q, --quiet, --silent

Do not output messages to stdout

-r, --mdm-repository

Collect MDM repository files

-s, --skip-sdbg

Skip collection of SDBG output

-S, --pause-core-generation

Pause core generation of PowerFlex components during data collection

--tech

Include technician option in help message

-u USERNAME, --username=USERNAME

Use USERNAME for PowerFlex login (default: scli default)

-w WORK_DIR, --work-dir=WORK_DIR

Use directory `WORK_DIR` for temporary files (default: /tmp)

-x FILE, --output-file=FILE

Store collected bundle as FILE (default: getInfoDump). Appropriate file name suffix (.tgz, .zip) is added automatically

If FILE is '-', write bundle to standard output (implies --quiet)

-z, --zip

Use ZIP format for the collected bundle

-h, --help

Show this help message and exit

Related information

Collecting debug information using get_info
Retrieve the PowerFlex core component logs
Collect PowerFlex management platform support data
Collect PowerFlex Manager platform installer support data

Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform install problems.

1. If required, transfer the data collection executable, `pfmp_support` to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site. To get the utility file from the PowerFlex management platform installer, go to the `PFMP_Installer/scripts` directory. To download the data collection utility from the support site, search for **PFMP log collection Script** file on the Dell Technologies Support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, execute the data collection utility as a user with superuser permissions, such as user root.

   In the following example, the utility's executable exists under `/root`.

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, execute the utility as a user with superuser permissions, such as user root, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
```

```
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Related information

Collecting debug information using get_info

Collect PowerFlex Manager platform installer support data

Use the following procedures to collect the PowerFlex Manager platform installer support data.

1. On the PowerFlex Manager Platform (PFMP) installer host, execute the data collection utility, `pfmp_support`, as a user with superuser permissions, such as user root.
   The utility is available in the location where the installer was extracted, under the `PFMP_Installer/scripts` directory.
   NOTE: The PowerFlex management platform installer is a set of scripts and installer containers used to deploy the PowerFlex management platform and the PowerFlex Manager applications that run on the platform.
   In the following example, the PFMP installer was extracted under `/tmp/PFMP2-4.0.0-161`.

```
# /tmp/PFMP2-4.0.0-161/PFMP_Installer/scripts/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

2. Provide the resulting support bundle file to PowerFlex Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Related information

Collecting debug information using get_info

Generating a troubleshooting bundle

A troubleshooting bundle is a compressed file that contains logging information for PowerFlex Manager managed components. If necessary, download the bundle and send it to Dell Technologies Support for issue debugging.

The troubleshooting bundle includes the following logs:

- ASM deployer
- iDRAC lifecycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods

- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes
- PowerFlex Block logs

NOTE: You can generate troubleshooting bundles from the **Resource Groups** page too.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Generate Troubleshooting Bundle**.
3. In **PowerFlex Node Selection**, choose one of the following logs:
   - Logs from all PowerFlex nodes
   - Logs from specified PowerFlex nodes

   If you select the **Logs from specified PowerFlex nodes** option and select the number of nodes for which you want to generate the log, the **View/Select Nodes** button is displayed. Click the button to view the list of nodes in the **Node Selection** window. Select the required nodes from the **Available Nodes** list and click **>>** to view them in the **Selected Nodes** list. Click **Save** to return to the **Generate Troubleshooting bundle** page.

   For the **Logs from specified PowerFlex nodes** option, **Generate** is enabled only if a node is selected in the **Node Selection** window.
4. Select a **PowerFlex Log Level**:
   - Default node logs
   - Default node logs plus additional MDM information
   - Latest logs only
5. Optionally, click the checkbox **Include VMware ESXi logs** for all VMware ESXi logs.
6. Choose a **Delivery Option**:

Option	Description
Send to Dell Technologies connectivity services	If you are using Dell Technologies connectivity services, **Send to Dell Technologies connectivity services** is selected by default. Leave this default setting. If Dell Technologies connectivity services is not configured, this option is disabled.
Download to shared file path	Provide a path using the following format with credentials: `CIFS:\\1.1.1.1\uploadDirectory`. Enter a username and password, and click **Test Connection** to verify the connection to the CIFS share before generating the bundle.
Download locally	Download the troubleshooting bundle to a local file. When the troubleshooting bundle is finished, click **Download File** to save to a local drive.

7. Click **Generate**.

Sometimes, the troubleshooting bundle does not include log information for all the nodes. The log collection may appear to succeed, but the log for one or more of the nodes may be missing. You may see an error message in the scaleio.trace.log file that says `Could not run get_info script`. If you see this message, you may need to generate the troubleshooting bundle again to include information for all the logs.

Retrieve PowerFlex core component logs using REST API

Use the following procedures to collect logs using REST API.

- The PowerFlex management platform (PFMP) cluster must be present.
- The PowerFlex cluster must be deployed.
- The PowerFlex management platform cluster must be able to communicate with any host instance in the PowerFlex cluster. To test the connection, ping from a PowerFlex node to the data IP addresses of the instances.
- Ensure that you have mdm_ips, mno_username, mno_password, and mno_ip.

1. To log in to M&O, run the following API command:

   `POST https://<mno_ip>/rest/auth/login`

   For example:

```
token=$(curl -k --silent 'https://<mno_ip>/rest/auth/login' --header 'Accept:
application/json' --header 'Content-Type: application/json' -d '{"username":
"<mno_username>","password": "<mno_password>"}' |jq -r '.access_token'); echo $token
```

   Save the access token received in the output.

2. To get configuration, run the following API command:

   `POST https://<mno_ip>/im/types/Configuration/instances`

   For example:

   ```
   curl --silent  -k -X POST  -H "Authorization: Bearer ${token}" -H
   'Content-Type: application/json' -d '{ "mdmIps":["10.234.177.103","10.234.177.153"],
   "mdmUser":<mno_username>, "mdmPassword":"<mno_password>",
   "securityConfiguration":{ "allowNonSecureCommunicationWithMdm":"false",
   "allowNonSecureCommunicationWithLia":"false", "disableNonMgmtComponentsAuth":"false"
   } }'  https://<mno_ip>/im/types/Configuration/instances>config.json
   ```

   Save the output of the request. This output is a JSON representation of the system configuration.

3. Add the M&O login information to the JSON. The get info script requires the login information to perform certain queries, to provide this information, add the following key value pairs to JSON:

   ```
   mnoUser : "<mno_username>"
   mnoPassword : "<mno_password>"
   mnoIp : "<mno_ip>"
   ```

   For example:

   ```
   {
   "mnoUser" : "<mno_username>" ,
   "mnoPassword": "<mno_password>" ,
   "mnoIp: "<mno_ip>"
   "snmpIp": null
   … (rest of the json)
   }
   ```

4. To collect logs, run the following API command:

   `POST https://<mno_ip>/im/types/NodeInfo/instances/actions/collectLogs`

   To run the collect logs request, ensure that you have a valid token and the configuration JSON. For more optional attributes, go to step 7.

   ```
   curl -k -X POST  -H "Authorization: Bearer ${token}" -H 'Content-Type:
   application/json' -d @config.json  https://<mno_ip> /im/types/NodeInfo/instances/
   actions/collectLogs
   ```

5. To monitor the log collection process, run the following API command:

   `GET https://<mno_ip>/im/types/ProcessPhase/actions/queryPhaseState`

   Monitor the `phaseStatus` value and wait until the operation is complete.

   For example:

   ```
   curl -s -k  -X GET -H "Content-Type:application/json" -H "Authorization: Bearer $
   {token}" https://<mno_ip> /im/types/ProcessPhase/actions/queryPhaseState
   ```

   Output example:

   ```
   {"phaseStatus":"completed","phase":"query","numberOfRunningCommands":0,"numberOfPendin
   gCommands":0,"numberOfCompletedCommands":6,"numberOfAbortedCommands":0,"numberOfFailed
   Commands":0,"failedCommand
   ```

6. To download the logs, run the following API command:

   `GET https://<mno_ip>/im/types/NodeInfo/instances`

   For example:

   ```
   curl -s -k -i -X  GET -H "Content-Type:application/json" -H "Authorization: Bearer
   ${token}" https://<mno_ip>/im/types/NodeInfo/instances -o "get_info.zip"
   ```

   In this example, the logs are downloaded to the current working directory under the name "get_info.zip".

7. (Optional) Add the following optional attributes to the log collection request as query parameters:

- `targetIPs`—Filter the result to only part of the nodes (by their IP addresses)
- `copyRepositories`—Copy MDM repositories (true/false, default is false)
- `liteVersion`(lite version)—Collect trc.0, exp.0, and umt.0 files only and repository files (true/false, default is false)
- `copyBinaries`—Collect MDM, SDS, SDR, SDT, LIA binaries and core dumps (true/false, default is false)
- `collectSdbgScreens` (true/false, default is false)

For example:

```
curl  -k -X -i POST -H "Content-Type:application/json" -H "Authorization:
Bearer ${token}"   "https://<mno_ip>/im/types/NodeInfo/instances/actions/collectLogs?
copyRepositories=true&targetIPs=10.55.118.61&targetIPs=10.55.118.62"
```

8. To clear the operation and move to idle, run the following API commands:

   `POST https://<mno_ip>/im/types/Command/instances/actions/clear`

   `https://<mno_ip>/im/types/ProcessPhase/actions/moveToIdlePhase`

   After the log collection operation is complete, mark the completed operation as completed to allow the other jobs to run.

   For example:

```
curl -s -k  -X POST -H "Content-Type:application/json" -d '{}' -H "Authorization:
Bearer ${token}" https://<mno_ip>/im/types/Command/instances/actions/clear
```

```
curl -s -k  -X POST -H "Content-Type:application/json" -d '{}' -H "Authorization:
Bearer ${token}" https://<mno_ip>/im/types/ProcessPhase/actions/moveToIdlePhase
```

Retrieving additional PowerFlex logs

This section contains instructions for retrieving RAID controller, vSphere PowerFlex plug-in, PowerFlex Installer, and system event logs.

Retrieve RAID controller logs from systems

Perform this procedure to retrieve the RAID controller logs from ESXi- based and Linux-based servers in a system.

Ensure that you have access to the following:

- IP address of the node
- Password for the root user in ESXi-based and Linux-based systems
- The RAID controller utility, which is enabled (for details, see the *Hardware Configuration and Operating System Installation Guide* of your system)

1. SSH or RDP to the node.
2. Retrieve the controller information:

Operating System	Run this command

ESXi	`/opt/lsi/perccli/perccli /call show all > <file_name>`
	Example:
	`/opt/lsi/perccli/perccli /call show all > /var/tmp/store/RAIDinfo.txt`

Linux	`/opt/MegaRAID/perccli/perccli64 /call show all > <file_name>`
	Example:
	`/opt/MegaRAID/perccli/perccli64 /call show all > /var/tmp/store/RAIDinfo.txt`

3. Retrieve the `Show Events` log file:

Operating System	Run this command

ESXi	`/opt/lsi/perccli/perccli /call show events file=<file_name>`
	Example:
	`/opt/lsi/perccli/perccli /call show events file=/var/tmp/store/RAIDevents.txt`

Linux	`/opt/MegaRAID/perccli/perccli64 /call show events file=<file_name>`
	Example:
	`/opt/MegaRAID/perccli/perccli64 /call show events file=/var/tmp/store/RAIDevents.txt`

You can retrieve the `RAIDevents.txt` file from your local drive.

4. Retrieve the `Termlog` log file:

Operating System	Run this command

ESXi	`/opt/lsi/perccli/perccli /call show termlog file=<file_name>`
	Example:
	`/opt/lsi/perccli/perccli /call show termlog file=/var/tmp/store/RAIDtermlog.txt`

Linux	`/opt/MegaRAID/perccli/perccli64 /call show termlog file=<file_name>`
	Example:
	`/opt/MegaRAID/perccli/perccli64 /call show termlog file=/var/tmp/store/RAIDtermlog.txt`

You can retrieve the `RAIDtermlog.txt` file from your local drive.

Retrieve system event logs in VxFlex Ready Node servers

Perform this procedure to access the system events logs (SEL) in a VxFlex Ready Node server.

Ensure that you have access to the following:

- IP address of the iDRAC port
- Login credentials for the iDRAC (admin/admin as default username/password)

1. From a web browser, go to **`http://<IP_address_iDRAC_port>`**.
2. In the **Console Login** window, type the user name and password, then click **Login**.
3. To view the event log, select **View Logs** in the **Quick Launch Tasks** pane.
   The **System Event Log** is displayed with color-coded severity levels.
4. To save the event log, click the **Save As** button at the bottom of the table.
   The event log is saved in a `.CSV` file.

   The power supply- and fan-related errors in **Event Log** may be intermittent in nature, and therefore may display repetitive events for one module. For such errors, it is recommended that you remove the module and replace it in its socket. This might be a connection-related issue.

Collect logs from the management VM cluster

Perform the following procedure to collect logs from the management VM cluster.

1. SSH as root to the management installer node.
2. Run the following command: `/opt/dell/pfmp/PFMP_Installer/scripts/pfmp_support`

```
installer-30:/opt/dell/pfmp/PFMP_Installer/scripts # ./pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

12

Enabling audit logging

This section describes how to enable audit logging in PowerFlex Manager.

The procedures in this section explain how to enable logging for PowerFlex events and Ingress audit messages so this information can be forwarded to an external Security Information and Event Manager (SIEM).

Define the PowerFlex events notification policy

Use this procedure to define a notification policy to forward events in the PowerFlex system to the rsyslog-forwarder (also known as the syslog-listener). Then, the rsyslog-forwarder forwards the events to the external destinations that are defined in the policy.

For this policy, you do not need to define a source, since the required source for PowerFlex events is a built-in feature.

1. Add a destination:

   First, you must add the identified Security Information and Event Manager (SIEM) server as a destination.

   a. Go to **Settings** > **Events and Alerts** > **Notification Policies**.

      You can also use the following REST API: `dispatch-destinations/post`

   b. From the **Destinations** pane, click **Add**.
      The **Create New Destination Protocol** window opens.

   c. Enter the destination name and description.

   d. From the **Destination Type** menu, select **Syslog**.

   e. Click **Next** and enter the IP, port, and protocol (TCP) of the target SIEM. Ensure that the SIEM IP, port, and protocol are reachable.

2. Create a new policy:

   The new policy defines the rules for processing PowerFlex event messages from sources and specifies to which destination that information should be sent.

   a. Go to **Settings** > **Events and Alerts** > **Notification Policies**.

      You can also use the following REST API: `dispatch-policies/post`

   b. Click **Create New Policy**.

   c. Enter a name and a description for the notification policy. For the policy name, you can enter: **`Powerflex events to external Syslog`**

   d. Set the **Source Type** to **Powerflex_events**.

   e. From the **Resource Domain** menu, select the resource domain for the notification policy. The resource domain options are:

      - All
      - Management
      - Block (Storage)
      - File (Storage)
      - Compute (Servers, Operating Systems, virtualization)
      - Network (Switches, connectivity etc.)
      - Security (RBAC, certificates, CloudLink etc)

   f. Select the check box beside the severity levels that you want to associate with this policy.

      The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.

   g. Select the destination that is created in the previous step and click **Submit**.

Define the Ingress notification policy

Use this procedure to define a notification policy to forward Ingress audit messages. The purpose of this policy is to capture POST, PUT, and DELETE requests of signed-in users passing through the Ingress Controller and send them to the rsyslog-forwarder (also known as the syslog-listener). The rsyslog-forwarder then forwards them to the external destinations.

1. Add a syslog source, if you do not have one already:
   a. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
      You can also use the following REST API: `dispatch-sources/post`
   b. From the **Sources** pane, click **Add**.
      The **Add Source** window opens.
   c. Enter a source name and description. For the name, you can enter **`Ingress`**.
   d. For the type, select **Syslog** and click **Enable Syslog**.
2. Create a destination, if you do not have one already.
   You can use the destination that you created to define the event to syslog audit notification policy from the previous procedure.
3. Create a new policy:
   a. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
      You can also use the following REST API: `dispatch-policies/post`
   b. Click **Create New Policy**.
   c. Enter a name and a description for the notification policy. For the policy name, you can enter: **`Powerflex Ingress to external Syslog`**
   d. Set the **Source Type** to **Syslog**.
   e. Set the **Facility** to **Auditlog**.
   f. Select the check box beside the severity levels that you want to associate with this policy.
      The severity indicates the risk (if any) to the system, in relation to the changes that generated the audit messages.
   g. Select the destination that is created in the previous step and click **Submit**.

Change Ingress setting to emit audit messages

Use this procedure to ensure that requests of signed-in users are sent.

This procedure provides the steps that are required to turn the silent flag off (set it to false) in the Ingress audit plug-in. If this flag is not turned off, the audit messages that are related to POST, PUT, and DELETE requests of signed-in users passing though the Ingress Controller will not be sent.

1. Log in to one of the management and orchestration cluster nodes with kubectl permissions. Then, copy and paste the following script for enabling or disabling the silent flag:

```
kubectl get cm -n kube-system lua-audit-conf -o yaml > ./lua-audit-conf.current.yaml
sed 's/"silent": true/"silent": false/g'  ./lua-audit-conf.current.yaml > ./lua-audit-
conf.silent-off.yaml
sed 's/"silent": false/"silent": true/g'  ./lua-audit-conf.current.yaml > ./lua-audit-
conf.silent-on.yaml
echo "Copy + Paste either:"
echo "kubectl apply -f ./lua-audit-conf.silent-on.yaml --force"
echo "OR:"
echo "kubectl apply -f ./lua-audit-conf.silent-off.yaml --force"
```

2. Run the following command to start the audit:

```
kubectl apply -f ./lua-audit-conf.silent-off.yaml --force
```

3. Wait one minute.
4. To ensure the configuration changes made are reflected across all nodes, you must now refresh the user interface. To refresh the user interface, quickly press **F5** or **Ctrl+R** on the browser while on the user interface five times.
   All Ingress audit messages are now forwarded to the configured SIEM servers.

13

NVMe over TCP

PowerFlex administrators can implement NVMe over TCP and use NVMe over TCP features.

NVMe over TCP load balancing

PowerFlex supports load balancing with NVMe over TCP. NVMe over TCP load balancing requires persistent discovery and host network awareness. You must also add NVMe targets and create a system data network.

Implementing NVMe over TCP load balancing workflow

The following steps summarize the workflow that you must follow to use NVMe over TCP load balancing:

1. Set up persistent discovery on the host operating system.
2. Add NVMe targets.
3. Create a subnet.
4. Create a system data network.

Related concepts

Persistent discovery
NVMe targets
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

Persistent discovery

The persistent discovery controller ensures that the host remains connected to the discovery service after discovery. If there is a change in the discovery information that is provided to the host, the discovery controller returns an asynchronous event notification (AEN). The host requests the updated **Discovery log** page.

Here are some examples of changes in discovery information:

- A new volume is mapped to the host from a new protection domain.
- A new storage data target (SDT) is added to the system.
- Load balancing wants to move the host connection from one storage data target to another.

NOTE: To use persistent discovery, ensure the following:

- The host operating system supports the Persistent Discovery Controller
- Every host is connected to a subnet to enable discovery
- The persistent flag is used in the NVMe discover command

See the host operating system's documentation for NVMe over TCP host configuration instructions.

Related concepts

NVMe over TCP load balancing
NVMe targets
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

NVMe targets

NVMe targets (or SDT components) must be configured on the PowerFlex system side, in order to use NVMe over TCP technology.

The NVMe target (or SDT component) is a frontend component that translates NVMe over TCP protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.

TCP ports, IP addresses, and IP address roles must be configured for each NVMe target (or SDT component). You can assign both storage and host roles to the same target IP addresses. Alternatively, assign the storage role to one target IP address, and add another target IP address for the host role. Both roles must be configured on each NVMe target.

- The host port listens for incoming connections from hosts, over the NVMe protocol.
- The storage port listens for connections from the MDM.

Once the NVMe targets have been configured, add hosts to PowerFlex, and then map volumes to the hosts. Connect hosts to NVMe targets, preferably using the discovery feature.

On the operating system of the compute nodes, NVMe initiators must be configured. Network connectivity is required between the NVMe targets and the NVMe initiators, and between NVMe targets (or SDT components) and storage nodes.

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

Related tasks

Add an NVMe target
Modify an NVMe target
Remove an NVMe target

Add an NVMe target

Add an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. Click **Add NVMe Target**.
3. From the **Add NVMe Target** dialog box, configure the following settings:
   a. For **Target Name**, enter a name for the NVMe target (or SDT component).
   b. For **Protection Domain**, select a protection domain.
   c. Accept the default TCP ports, or modify them if necessary.
      - **Storage Port** listens for incoming connections from the MDM, and is only open on an IP address configured with the storage role. Default=12200
      - **I/O Port** listens for incoming connections for I/O, and is only open on an IP address configured with the host role. Default=4420 (which conforms to the NVMe over TCP standard)
      - **Discovery Port** listens for incoming discovery connections, and is only open on an IP address configured with the host role. Default=8009 (which conforms to the NVMe over TCP standard. Use zero to disable the discovery service.)
   d. For **Target IPs**, enter an IP address, and then select a role for the IP address from the **Target IPs Roles** menu. Click **Add Target IP**, and repeat this step until all IP addresses are configured.
4. Click **Add**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Related concepts

NVMe targets

Related tasks

Modify an NVMe target
Remove an NVMe target

Modify an NVMe target

Modify the configuration of an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Modify**.
3. In the **Modify NVMe Target** dialog box, modify the desired fields.
4. Optionally, add more target IP addresses, by clicking **Add Target IP**.
5. Click **Modify NVMe Target**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Related concepts

NVMe targets

Related tasks

Add an NVMe target
Remove an NVMe target

Remove an NVMe target

Remove an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Remove**.
3. In the **Remove NVMe Target** dialog box, verify that you are removing the wanted NVMe target, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Related concepts

NVMe targets

Related tasks

Add an NVMe target
Modify an NVMe target

NVMe over TCP host network awareness

Hosts are connected to the storage through a Layer 2 or Layer 3 network. While the storage does not manage the network and is not aware of the network configuration, the storage is impacted by the following network aspects:

- Load balancing takes the data network or subnet into consideration to ensure that there is a balance between the host connections on each data network with a specific subnet for improved performance.
- In Layer 3 networks, the PowerFlex system must have configured routing tables.

Two network objects must be defined for load balancing to work:

Host subnet	Networks or subnets that are used to connect hosts to storage. A subnet is also used to support static routes for data networks. These networks or subnets can be either Layer 2 or Layer 3. Layer 3 networks require configured routing tables. **NOTE:** When configuring NVMe hosts, ensure that every host is connected to a subnet to allow for discovery.

System data network	System-wide object that applies to all protection domains. After a resource group is deployed, you must configure system data networks or subnets to be used to connect the hosts initiator to the storage data target. The number of system data network that you need depends on the number of PowerFlex data networks that are configured for the deployment. PowerFlex data networks are used for storage node to storage node and storage node to SDT communication. You may have to define the system data networks before starting the PowerFlex deployment. **NOTE:** PowerFlex supports a maximum of eight system data networks.

If you have not defined the system data network, by default, a host can reach all system data networks. Not defining the system data networks or subnets may cause the following problems:

- Unequal load between the host initiator ports and nonoptimized I/O performance with the PowerFlex system.
- Impacted path resiliency if some of the host initiator ports cannot connect to the PowerFlex system.

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe targets
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

Managing system data networks in PowerFlex Manager

The NVMe over TCP load balancer uses the data networks for balancing the host connections. This section summarizes the steps that you must perform in PowerFlex Manager to add, rename, or remove system data networks.

Table 5. Add, rename, or remove data networks in PowerFlex Manager

If you want to...	Do this in PowerFlex Manager
Add system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Enter the required information and click **Save**.
Rename system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Select the network and click **Rename**. 4. Rename the network and click **Apply**.
Remove system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Select the network and click **Remove**. 4. Click **Remove** to remove the network.

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe targets
NVMe over TCP host network awareness
Managing system data networks using SCLI

Managing system data networks using SCLI

This section provides the SCLI commands that you need to perform to add, rename, or remove system data networks.

For more information on the PowerFlex SCLI commands, see *Dell PowerFlex CLI Reference*.

Table 6. Using SCLI to manage data networks

If you want to...	Use this SCLI command
Add system data networks	`scli --add_system_network --network_ip <IP> --network_mask <IP> [--network_name <NAME>] (--host_group_id <ID> | --host_group_name <NAME>)`
Rename system data networks	`scli --rename_system_network (--network_id <ID> | --network_name <NAME>) --new_name <NAME>`
Remove system data networks	`scli --remove_system_network (--network_id <ID> | --network_name <NAME>)`

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe targets
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing host groups using SCLI
Managing network sets using SCLI

Managing host groups using SCLI

This section provides the SCLI commands that you must perform to add, modify, rename, remove, or query host groups.

This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

The host group holds a subgroup of existing networks that are accessible to a group of hosts. Each host group is associated with a set of system networks. Each host may be associated with multiple host groups. By default, a host is part of the default host group until you create a new host group and associate the host to the new host group. By default, a host supports all data networks and you do not need to configure the host groups. The maximum number of host groups that are allowed per system is 1024.

Table 7. Managing host groups using SCLI

If you want to...	Use this SCLI command
Add host group	`scli --add_host_group [--host_group_name <NAME>] [(--network_id <ID> | --network_name <NAME>)] [(--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)]`
Modify host group	`scli --modify_host_group (--host_group_id <ID> | --host_group_name <NAME>) [(--network_id <ID> | --network_name <NAME>)]`
Rename host group	`scli --rename_host_group (--host_group_id <ID> | --host_group_name <NAME>) [--new_name <NAME>]`
Remove host group	`scli --remove_host_group (--host_group_id <ID> | --host_group_name <NAME>)`

Table 7. Managing host groups using SCLI (continued)

If you want to...	Use this SCLI command
Assign host to group	`scli --assign_host_to_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>) (--host_group_id <ID> | --host_group_name <NAME>)`
Remove host from group	`scli --remove_host_from_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)`
Query host group	`scli --query_host_group (--host_group_id <ID> | --host_group_name <NAME>)`
Query all host group	`scli --query_all_host_group`

Related concepts

Managing system data networks using SCLI
Managing network sets using SCLI

Managing network sets using SCLI

This section provides the SCLI commands that you need to perform to add, modify, rename, remove, or query network sets.

This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

A network set is a set of networks that are connected to the same switch. This switch is configured for balancing the I/O traffic between access switches by associating system data networks to network set. The load balancer takes care of the distribution of networks over network sets.

For example, if a system is defined with two system data networks (192.168.150.0 and 192.168.151.0), then you could create two network sets and associate 192.168.150.0 with network set 1 and 192.168.151.0 to network set 2.

Each system data network belongs to exactly one network set. If the network was not assigned to a network set, the network will be considered a network set of its own.

Use the default network set and do not manually create the network set. Creating network sets is applicable for a software-only deployment.

Table 8. Managing network sets using SCLI

If you want to...	Use this SCLI command
Add network set	`scli --add_network_set [--network_set_name <NAME>] [(--network_id <ID> | --network_name <NAME>)]`
Modify network set	`scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [(--network_id <ID> | --network_name <NAME>)]`
Rename network set	`scli --rename_network_set (--network_set_id <ID> | --network_set_name <NAME>) --new_name <NAME>`

Table 8. Managing network sets using SCLI (continued)

If you want to...	Use this SCLI command
Remove network set	`scli --remove_network_set (--network_set_id <ID> | --network_set_name <NAME>)`
Query network set	`scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)`
Query all network sets	`scli --query_all_network_set`

Related concepts

Managing system data networks using SCLI
Managing host groups using SCLI

Migrating to NVMe over TCP on VMware ESXi

Use the following guidelines when migrating a VMFS datastore and clusters from SDC to NVMe over TCP.

PowerFlex supports three options for migrating from SDC to NVMe over TCP on VMware ESXi:

Online migration using Storage vMotion (VMFS only)

This method is a standard way of using Storage vMotion to move storage and it also supports switching protocols by migrating to a new datastore. This is the method that is described in this guide.

Offline conversion (VMFS only)

Offline conversion is an option for converting an existing VMFS datastore from SCSI (SDC) to NVMe over TCP without having to copy all the data over the network. This option is covered in the Dell Knowledge Base article SCSI to NVMe VMware VMFS Datastore Offline Migration Steps.

Conversion for clustered applications using RDMs

This option converts existing SDC- and RDM-based application clusters to shared VMDKs on NVMe over TCP datastores. This option is covered in the Dell Knowledge Base article PowerFlex: SDC to NVMe/TCP conversion for clustered applications utilizing RDMs on vSphere.

NOTE:

- Dell Technologies recommends using VMware Storage vMotion to migrate data from a volume that is presented by SDC to one presented by NVMe over TCP.
- Linux environments are not discussed in this guide.

Requirements

See the following VMware vSphere 7.0 product documentation link for information about Storage vMotion requirements and limitations: Storage vMotion Requirements and Limitations.

See the following VMware vSphere 8.0 product documentation link for information about requirements and limitations of VMware NVMe Storage: Requirements for VMware NVMe Storage.

See the following VMware KB article for more details about storage migration (Storage vMotion), with the virtual machine powered on: Migrating virtual machines with Raw Device Mappings (RDMs).

Ensure that you satisfy these requirements before you begin the migration:

- The VMware ESXi version is 8.x or later.
- The PowerFlex version is 5.0 or later.

Workflow

The following steps summarize the workflow that you must follow to migrate from SDC to NVMe over TCP using Storage vMotion:

1. Create and map a new volume of equal or greater size than the current VMFS datastore using NVMe over TCP to the same host.
2. Scan for the newly mapped volume.
3. Create a datastore on the NVMe over TCP volume.
4. Perform the standard data migration using Storage vMotion by using a nondisruptive process.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Before migrating the data, you must prepare the VMware ESXi node for mapping NVMe over TCP volumes.

Ensure that you satisfy these requirements before you begin:

- You must have an NVMe over TCP target supported PowerFlex system (version 4.5.x or later).
- Deployed VMware ESXi compute-only nodes.
- The host must be at VMware ESXi version 7.x or later.

Related concepts

Migrating to NVMe over TCP on VMware ESXi

Related tasks

Enable the NVMe over TCP VMkernel ports
Add NVMe over TCP software storage adapter
Copy the host NQN
Add a host to PowerFlex
Create a volume
Map a volume to the host
Discover and connect the NVMe over TCP Target
Perform a rescan of the storage
Create a VMFS datastore on the NVMe over TCP volume

Enable the NVMe over TCP VMkernel ports

Use this procedure to enable the NVMe over TCP VMkernel ports.

1. Log in to the VMware vSphere Client.
2. Click **Home/Inventory** and select the host.
3. Select **Configure** > **VMkernel adapters**.
4. Edit **PowerFlex-Data 1**.
5. Select the **NVMe over TCP** check box and click **OK**.
6. Repeat these steps for the remaining PowerFlex data networks.

7. Repeat the steps for the remaining hosts in the cluster.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Add NVMe over TCP software storage adapter

Use this procedure to add an NVMe over TCP software storage adapter.

1. Log in to the VMware vSphere Client.
2. Click **Home** > **Inventory** > **Hosts and Clusters**.
3. Select the cluster of PowerFlex compute-only nodes that you want to configure.
4. From the right pane, click **Configure** > **Storage Adapters**.
5. From the right pane, click **Add Software Adapter**.
6. Click **Add NVMe over TCP adapter**.
7. Select the first VMNIC and click **OK**.
8. Click **Add NVMe over TCP adapter**.
9. Select the second VMNIC and click **OK**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Copy the host NQN

Use this procedure to copy the host NQN to the copy buffer. The host NQN details are required when you add the host to PowerFlex.

1. Log in to VMware vSphere Client.
2. Select the first host.
3. From the right pane, click **Configure** > **Storage Adapters**.
4. Select the first VMware NVMe over TCP storage adapter. For example, **vmhba6x**.
5. From the pane, select **Controllers/Add Controller**.

   The host NQN is listed at the top of the form.

6. Click **COPY** and place the host NQN in the copy buffer.
7. Click **CANCEL**.
8. Repeat the steps for all the hosts.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Add a host to PowerFlex

Use this procedure to add a host to PowerFlex.

1. Log in to PowerFlex Manager.
2. Click **Block** > **Hosts**.
3. Click **Add Host**.
4. Enter the hostname and paste the host NQN from the copy buffer.
5. Enter the **Number of Paths Per Volume**. The default number of paths is four per volume, and the maximum number of paths is eight per volume.
6. Enter the **Number of System Ports per Protection Domain**. The default number of system ports is ten per protection domain, and the maximum number of system ports is 16 per protection domain.
7. Click **Add**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Create a volume

Use this procedure to create a volume.

1. Log in to PowerFlex Manager,
2. Click **Block** > **Volumes**.
3. Click **Create Volume**.
4. Enter the number of volumes and the name of the volumes.
5. Enter the required volume size in GB, specifying the size in 1 GB increments.
6. Select the NVMe storage pool and click **Create**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Map a volume to the host

Use this procedure to map a volume to the host.

1. Log in to PowerFlex Manager,
2. Click **Block** > **Volumes**.
3. Select the volume check box and click **Provision** > **Map**.
4. Select the protocol **NVMe**.
5. Select the check box for the host to which you are mapping the volume.
6. Click **Map**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Discover and connect the NVMe over TCP Target

Use the procedure to discover and connect the NVMe over TCP Target PowerFlex system. Use the `esxcli` command to perform the operation.

1. Log in to the ESXi server using ssh.
2. Run the discovery query on each adapter:

   ```
   #esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
   #esxcli nvme fabrics discover -a vmhba6y -i 192.168.x.y -p 8009
   ```

   In the first example above, 6x is the first NVMe over TCP software adapter and 6y is the second NVMe over TCP software adapter.

   In the second example above, 192.168.x.x is first data IP Address and 192.168.x.y is second data IP Address, depends on which VMNIC is enabled for which software NVMe over TCP adapter.

3. Connect to the PowerFlex system by appending “-c” to the discovery query command.

   ```
   #esxcli nvme fabrics discover -a vmhba64 -i 192.168.x.x -p 8009 -c
   #esxcli nvme fabrics discover -a vmhba65 -i 192.168.x.y -p 8009 -c
   ```

4. Get the controller list by verifying the connected controllers.

   ```
   #esxcli nvme controller list
   ```

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Perform a rescan of the storage

Use this procedure to perform a rescan of the storage.

1. In the vSphere Client object navigator, browse to a host, a cluster, a data center, or a folder that contains hosts.
2. From the right-click menu, select **Storage** > **Rescan Storage**.
3. Specify the extent of the rescan.

Option	Description
Scan for New Storage Devices	Rescan all adapters to discover new storage devices. If new devices are discovered, they appear in the device list.
Scan for New VMFS Volumes	Rescan all storage devices to discover new datastores that have been added since the last scan. Any new datastores appear in the datastore list.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Create a VMFS datastore on the NVMe over TCP volume

Use this procedure to create a VMFS datastore on the NVMe over TCP volume.

1. Log in to VMware vCenter.
2. Select **Home/Inventory** and select the storage icon.
3. Right-click the cluster of PowerFlex compute-only nodes and select **Storage/New Datastore**.
4. Select VMFS and click **NEXT**.
5. Enter the name of the datastore and select a host from the list.
6. You should see a new disk called NVMe TCP Disk (`eui.######`).
7. Select the disk and click **NEXT**.
8. Select VMFS 6 and click **NEXT**.
9. Leave the default configuration unless you have been otherwise instructed and click **NEXT**.
10. Review your selections and click **FINISH**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes
Migrate the data with Storage vMotion

Migrate the data with Storage vMotion

After you prepared the node, you can migrate the data with Storage vMotion.

Follow the standard VMware vSphere 7.0 procedure to migrate the virtual machines from an SDC-based datastore to an NVMe over TCP-based datastore: Migrate a Virtual Machine to New Storage

Related tasks

Create a VMFS datastore on the NVMe over TCP volume